程序設計介紹

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一、程序設計內容

- 程序設計定義

所謂「程序」就是組合各種化學反應及物理變化的單元操作，將原料變為附加價值較高的產品之軟體及硬體，而「程序設計」即是設計這些單元操作流程的一件工作。

單元程序：加氫脫硫、烷化、氯化

單元操作：蒸餾、萃取、吸收、加熱、冷卻、流體輸送

軟體：操作條件、反應配方

硬體：反應器、蒸餾塔、熱交換器、貯槽、泵浦

實驗室

化工廠

產品

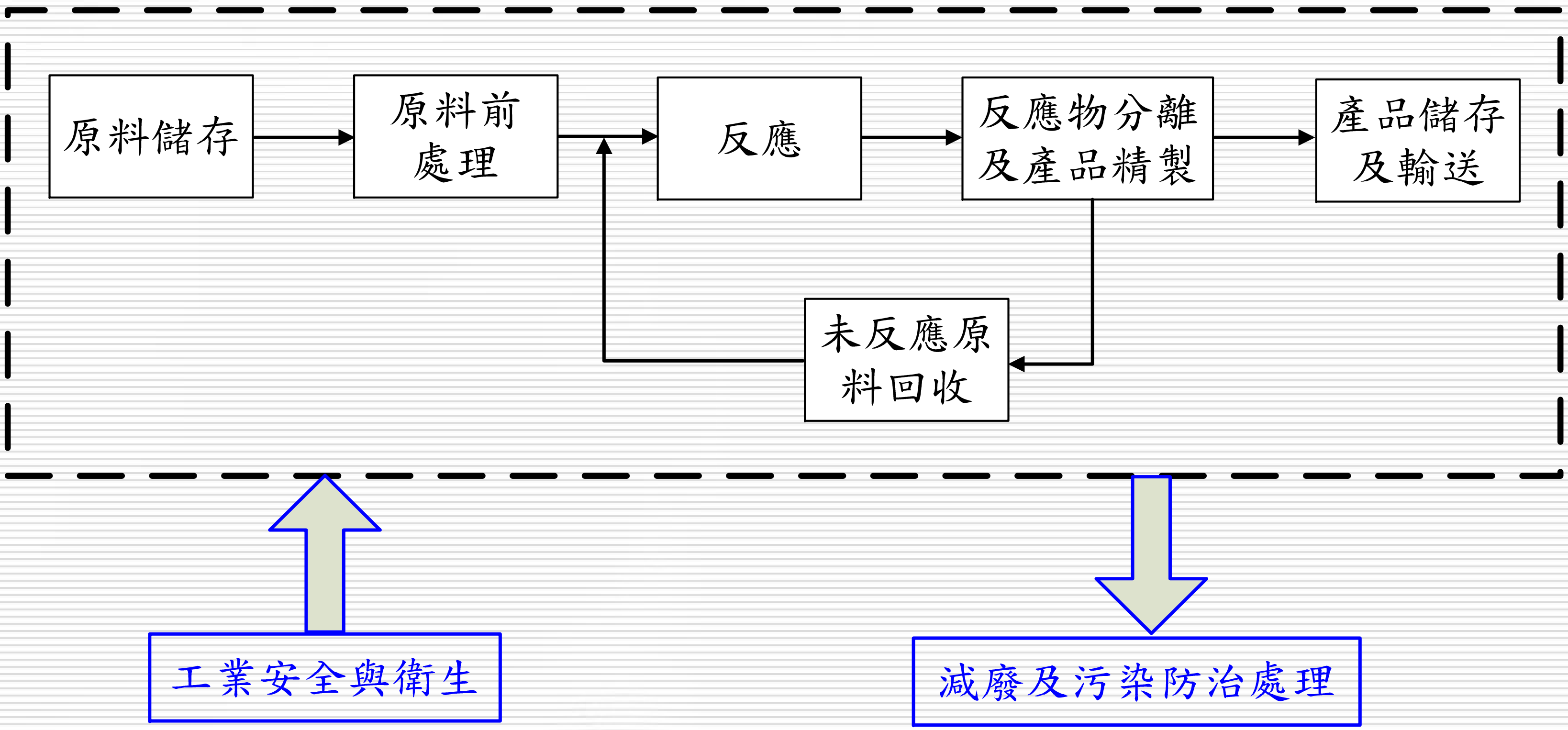
- 程序設計整合課程

  a. 質能平衡
  b. 單元操作
  c. 化工熱力學
  d. 反應工程
  e. 程序控制
  f. 工業安全
  g. 環境工程

- 程序設計應用場合

  a. 新廠設計
  b. 舊製程改善
  c. 產能去瓶頸

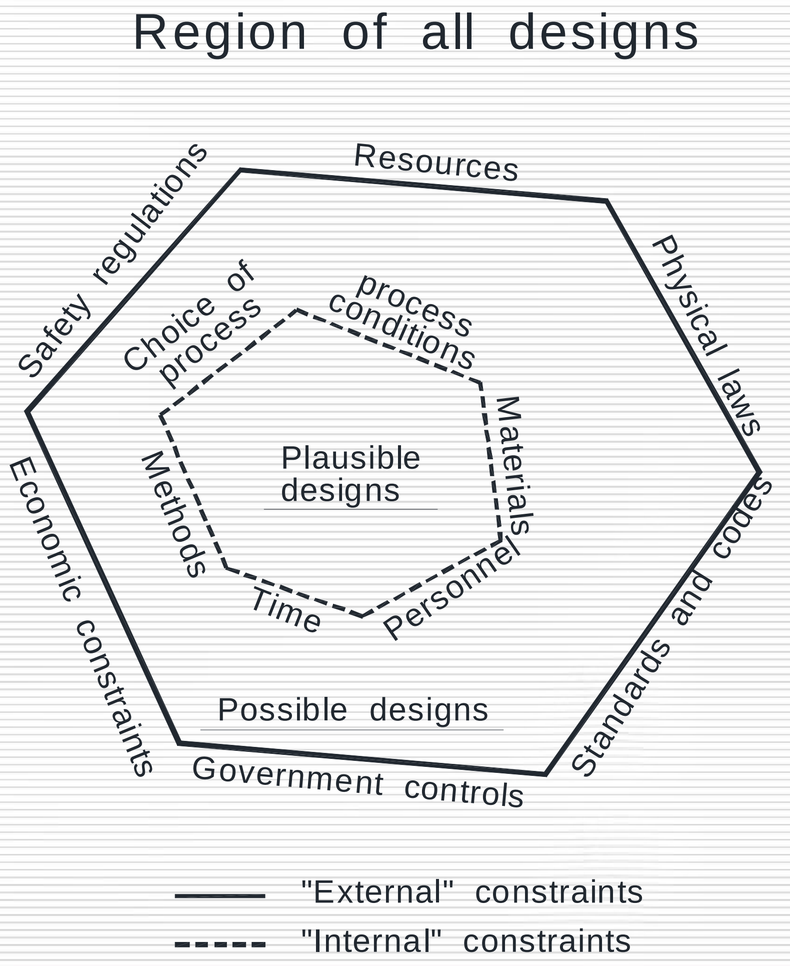
- 化工程序架構



製程危害分析、安全儀錶系統
<u>本質較安全設計</u>

製程優化減廢
<u>環保法規規範</u>

- 程序設計限制
  - 尋找可行且最適化的設計模式



- 基本工程設計數據 (BEDD)

  a. General matters

  - 專案說明、編號系統、法規、規範等...

  b. Design philosophy

  - 重要設備、真空設計條件、最小及最大可操作負載、密閉排放系統...

  c. Utility conditions

  - 公用流體的條件

	NORMAL		**MINIMUM**		MAXIMUM		DESIGN		Incremental
	P kg/cm^{2}G	*Temp.,sat'd* °C	**P** kg/cm^{2}G	***Temp.,sat'd*** °C	P kg/cm^{2}G	Temp. °C	P kg/cm^{2}G	Temp. °C	Value US$/T
S12	12	*191*	**11**	***187.2***	13	*195*	**16.0**	*230*	*30*
S1	1.5	*127*	**1.3**	***124.5***	2.4	*155*	3.0	*160*	*30*
			(Design basis for sizing)						

- 基本工程設計數據 (BEDD)(續)

  d. Site conditions

  - 廠址環境條件，如風力、溫度、濕度、地震強度、降雨量等…

  e. Environmental preservation

  - 廢氣、廢液等之排放標準

  f. Plant layout

  - 整廠規劃

  g. Basis for equipment and instrumentation

  - 設備及儀表之設計基準

  h. Other matters

- K/H & 基本設計內容

  Part (a) K/H package

  a. Design basis

     設計產能、原料及產品規格、原料及公用流體原單位消耗

  b. Process description

  c. Physical properties

  d. Process control flow sheet

  e. Material & energy balance

  f. Safety data

     製程安全危害分析、SIL等級等

  g. Preliminary equipment specification

  h. Operation guide

  i. Sample analysis procedure

  j. Man power requirement

- K/H & 基本設計內容(續)

Part (b) Basic design package

a. Basic engineering design data (BEDD)

b. Preliminary plot plan

c. Piping and instrumentation diagrams

d. Equipment specifications

e. General engineering specifications & Standard drawings

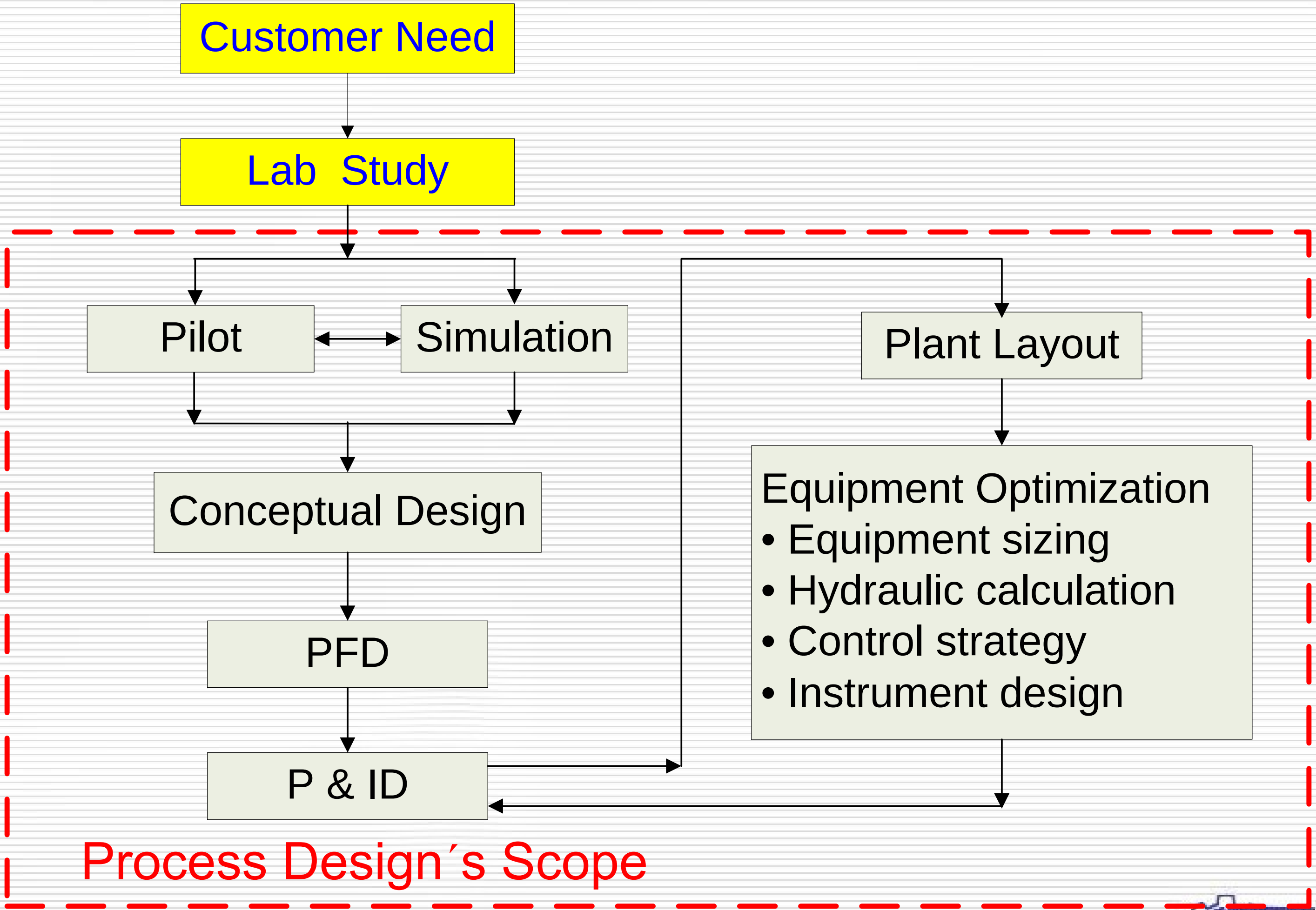
Equipment Specification中會用到的共用規範及標準圖

f. Piping and valve code

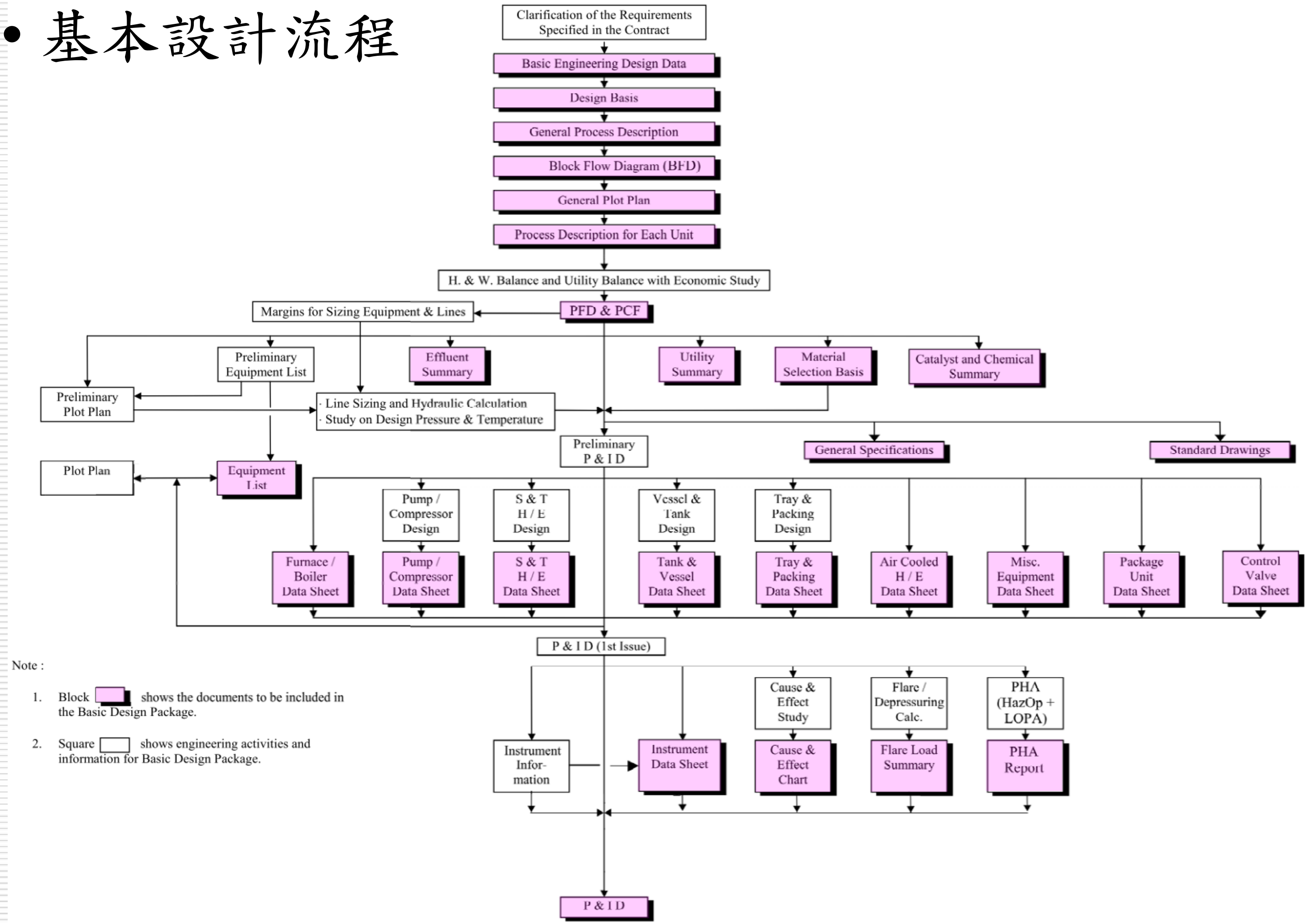
g. Instrument data list & specification

二、程序設計流程

- 程序開發流程



- 基本設計流程


Clarification of the Requirements Specified in the Contract
Basic Engineering Design Data
Design Basis
General Process Description
Block Flow Diagram (BFD)
General Plot Plan
Process Description for Each Unit
H. & W. Balance and Utility Balance with Economic Study
Margins for Sizing Equipment & Lines
PFD & PCF
Preliminary Equipment List
Effluent Summary
Utility Summary
Material Selection Basis
Catalyst and Chemical Summary
Preliminary Plot Plan
· Line Sizing and Hydraulic Calculation
· Study on Design Pressure & Temperature
Preliminary P & I D
General Specifications
Standard Drawings
Plot Plan
Equipment List
Pump / Compressor Design
S & T H / E Design
Vessel & Tank Design
Tray & Packing Design
Furnace / Boiler Data Sheet
Pump / Compressor Data Sheet
S & T H / E Data Sheet
Tank & Vessel Data Sheet
Tray & Packing Data Sheet
Air Cooled H / E Data Sheet
Misc. Equipment Data Sheet
Package Unit Data Sheet
Control Valve Data Sheet
P & I D (1st Issue)
Cause & Effect Study
Flare / Depressuring Calc.
PHA (HazOp + LOPA)
Instrument Infor-mation
Instrument Data Sheet
Cause & Effect Chart
Flare Load Summary
PHA Report
P & I D
Note :
1. Block shows the documents to be included in the Basic Design Package.
2. Square shows engineering activities and information for Basic Design Package.

- 製程模擬目的

a. 設計新廠或新製程

b. 檢討現有工廠操作

c. 獲得設備設計所須的操作條件及物性數據

d. 改善操作條件及節能

- 最佳化
- 產能放大
- 提高產品規格

- 製程模擬軟體之功能

a. 提供各種單元操作模組，包括:

穩態單元操作：蒸餾、吸收、氣提、萃取、驟沸、熱交換器、壓縮機、泵、加熱爐、Turbine、Expander、管線模擬器、貯槽、氣液分離槽、氣液液分離槽、控制器...

動態單元操作：動態蒸餾塔、動態反應器、動態緩衝槽、動態三相分相槽、換熱器、混合器、分流器、控制閥、PID控制器...

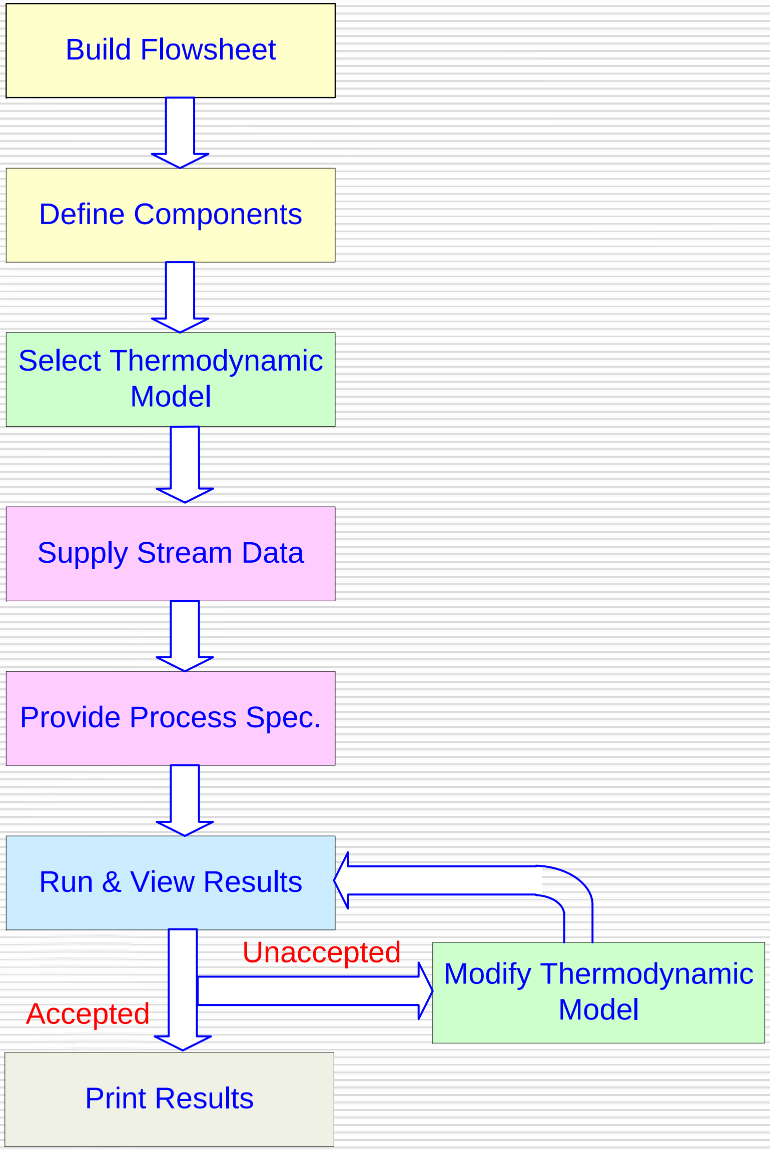
b. 提供物性資料庫及熱力學模式：以 AIChE 之 DIPPR 為基礎，加上電解質資料庫、相平衡模式、和熱焓模式和物性計算

c. 進行製程單元或整廠整合模擬、操作變數敏感度分析及最適化

d. 提供工程設計所需數據：蒸餾塔(板塔、填充塔)、吸收塔、熱交換器、貯槽、轉動機械、管線、控制閥及緊急排放閥....等設備Sizing

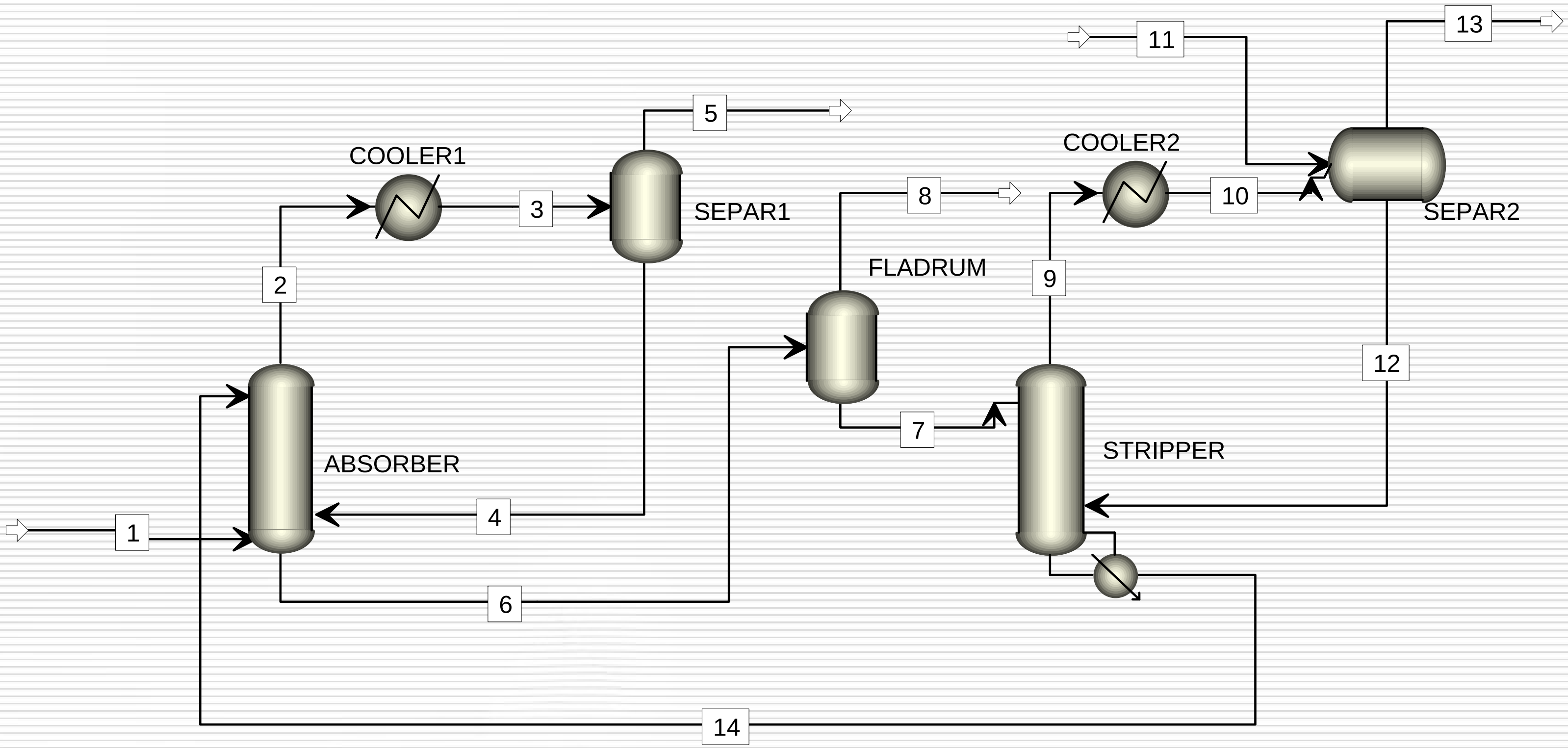
☆ 選擇『適合的』工具軟體，不一定要最貴的

- 製程模擬步驟

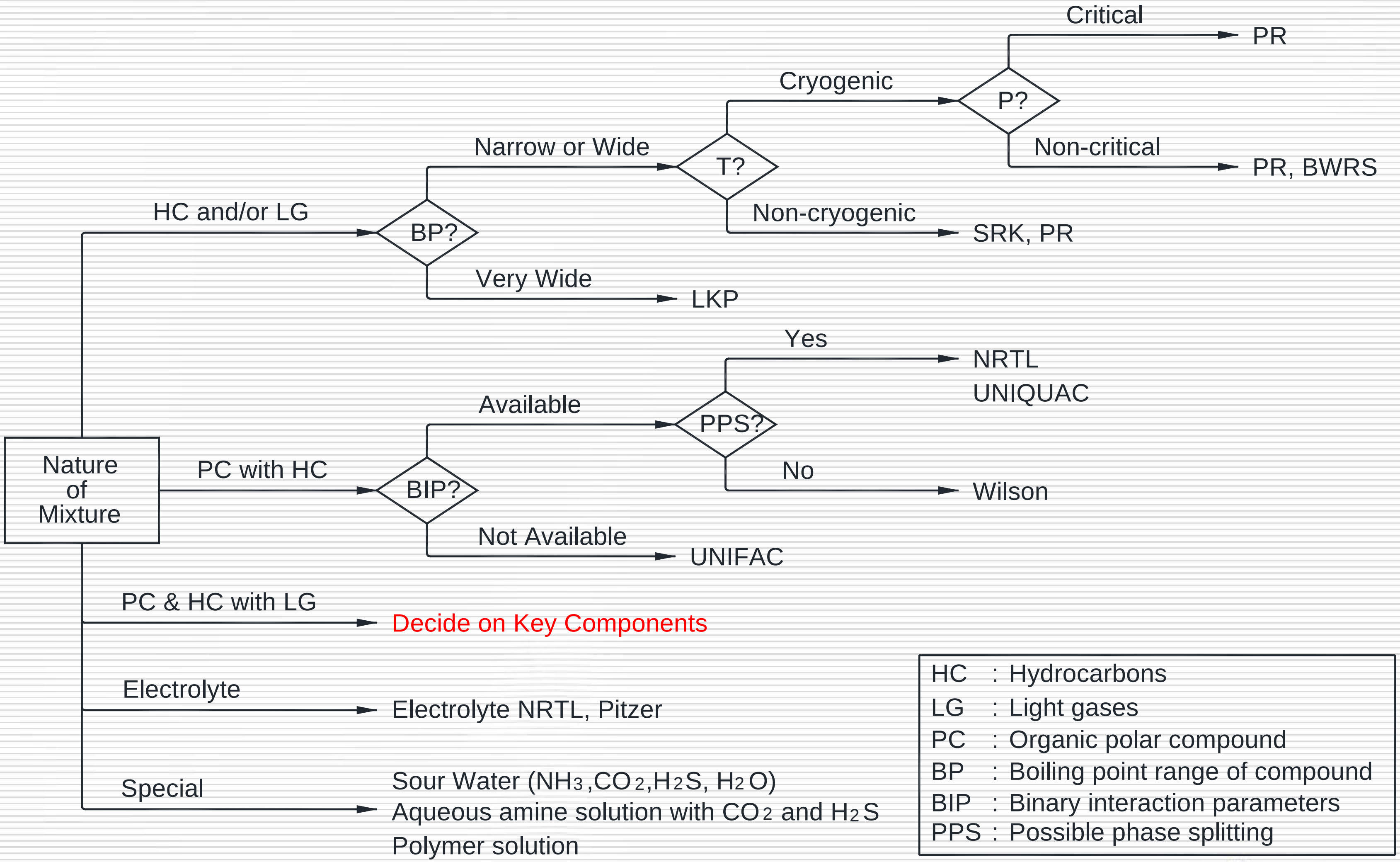

Build Flowsheet
Define Components
Select Thermodynamic Model
Supply Stream Data
Provide Process Spec.
Run & View Results
Unaccepted
Modify Thermodynamic Model
Accepted
Print Results

Build Flowsheet

Example for CO_2 Capture



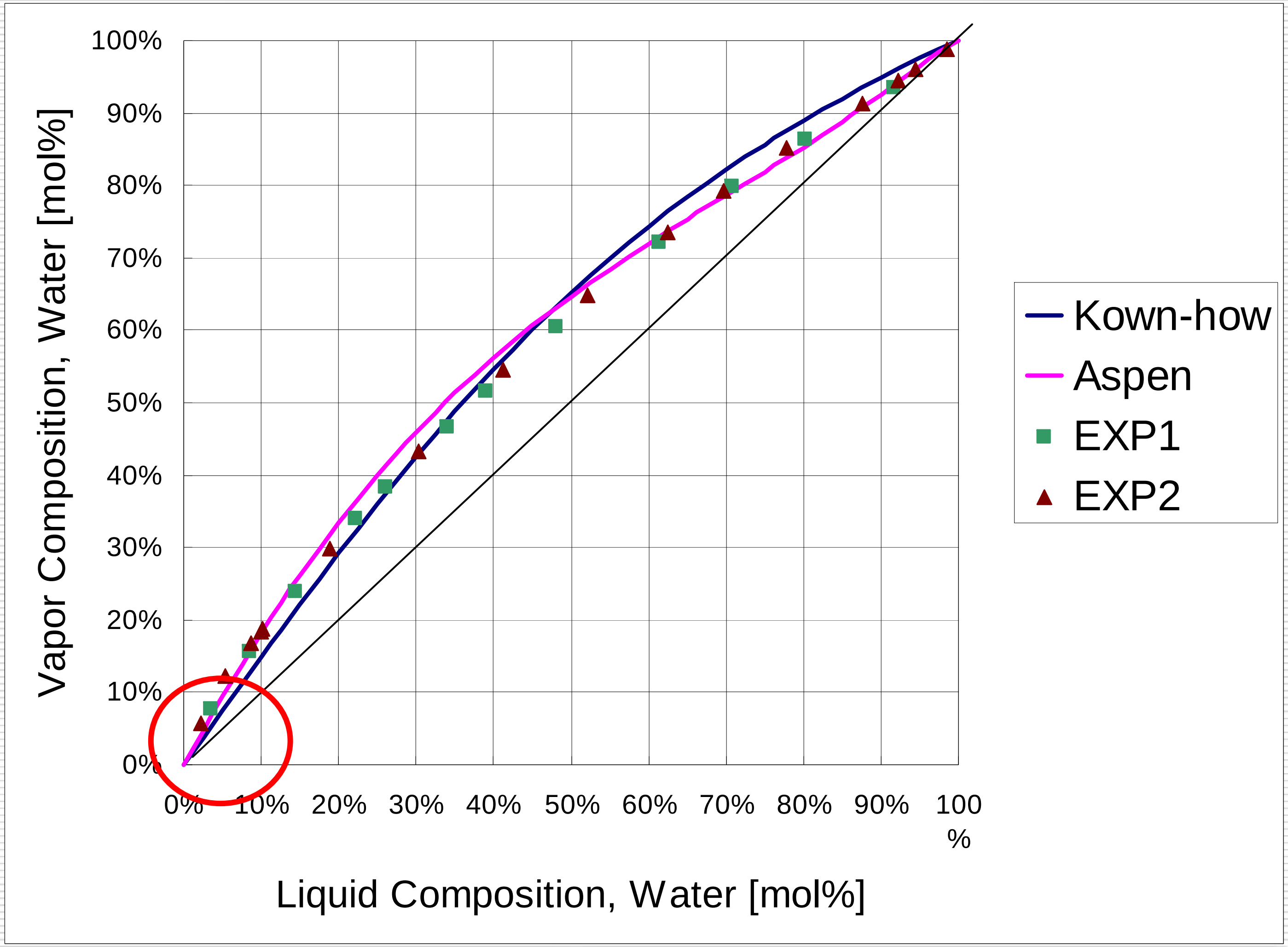
Select Thermodynamic Model



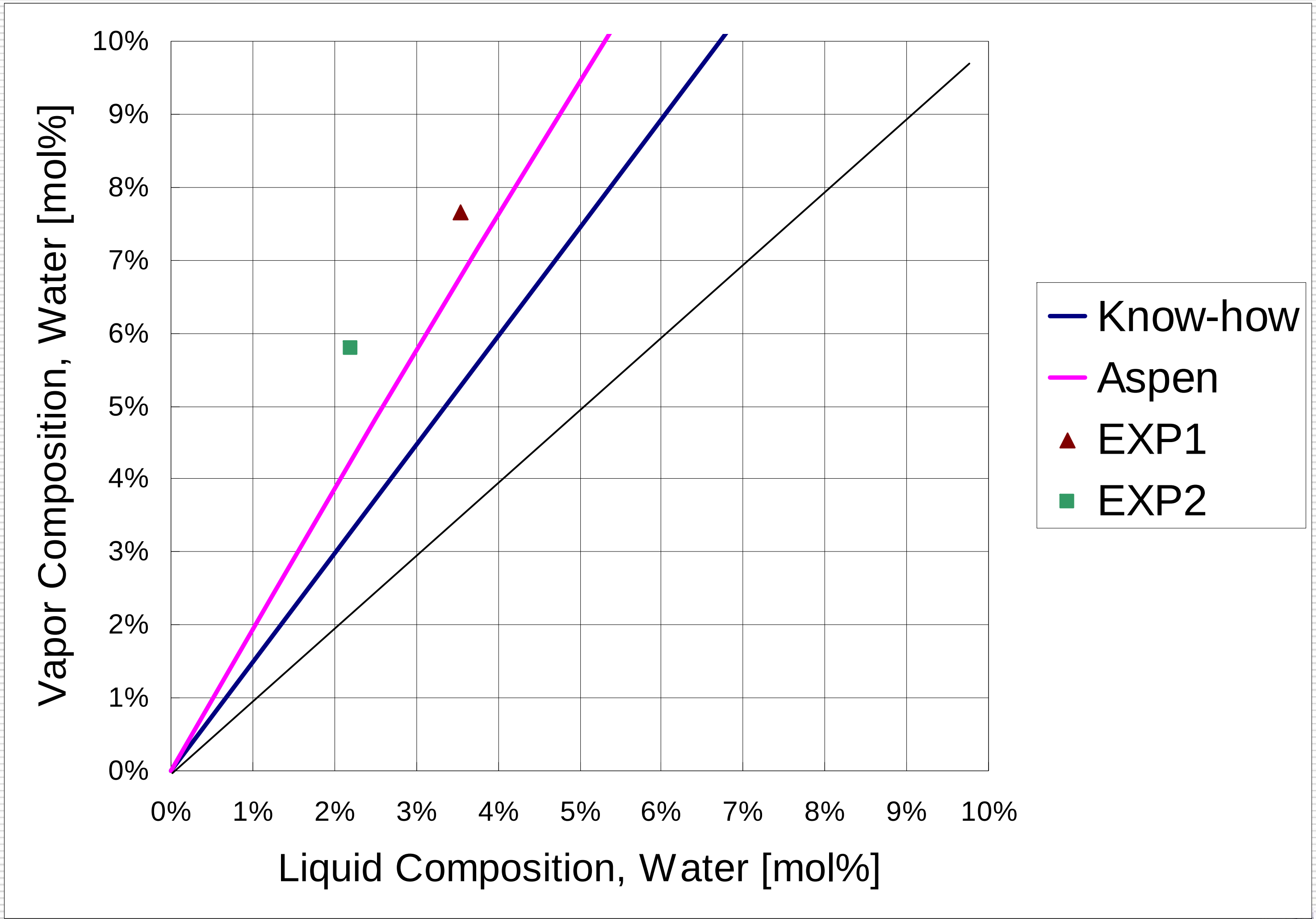
Thermodynamic Model Parameters Sources

a. Simulator data bank
b. DECHEMA or literatures
c. R & D data
d. Predictive model
    - UNIFAC, Regular Solution, Flory-Huggins Models
e. Simplified experimental data
    - Azeotropic data, mutual solubility, infinite dilution activity

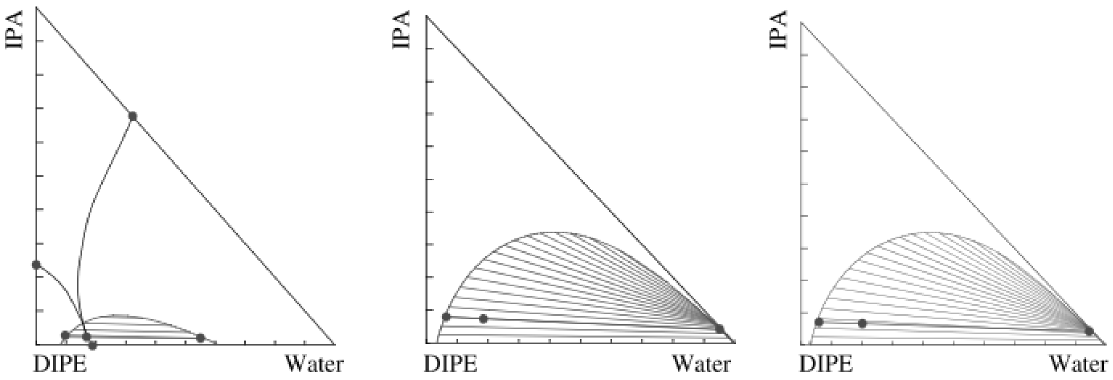
VLE Data Check



VLE Data Check (Water Removal Column)



LLE Data Check



(a) Decanter based on NRTL parameters correlated from VLE data at saturated condition.

(b) Decanter based on NRTL parameters correlated from LLE data at saturated condition.

(c) Decanter based on NRTL parameters correlated from LLE data with two-phase region plotted at 30°C.

Figure 12.34 The two-phase region can be plotted from correlations fitted to VLE or LLE data, and can be plotted at saturated conditions or at a fixed temperature.

- 流程圖發展

流程圖是利用各種符號（主要為設 備、管線及儀控等三類）把流程具體化、圖像化的圖件，依照設計時間的先後順序為:

a. Block Flow Diagram (方塊流程圖)

從原料進入至產品產出的製程中，主要的單元程序依照製程的順序表示的圖件

b. Process Flow Diagram (程序流程圖)

除了將單元程序的使用設備(含編號)表示出來外，圖中亦包括操作條件、流體流量及物性等資料

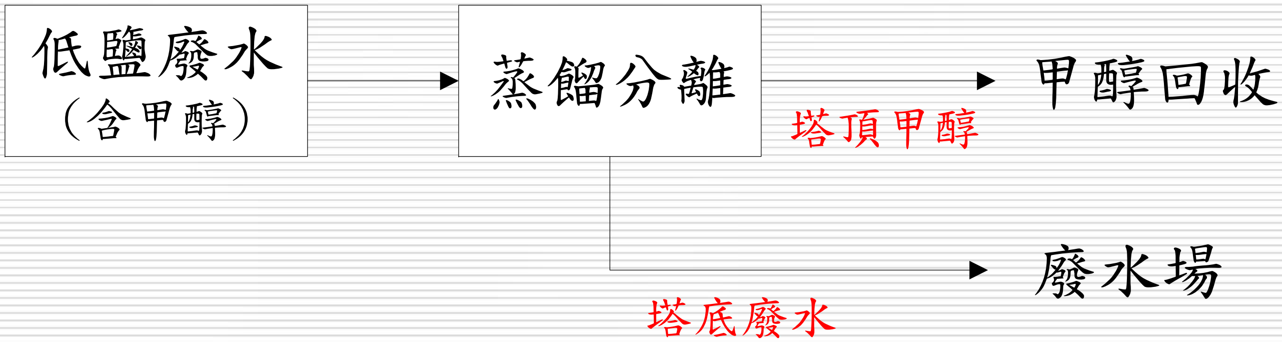
c. Process Control Flow Diagram (控制流程圖)

PFD中再增加流程控制迴路

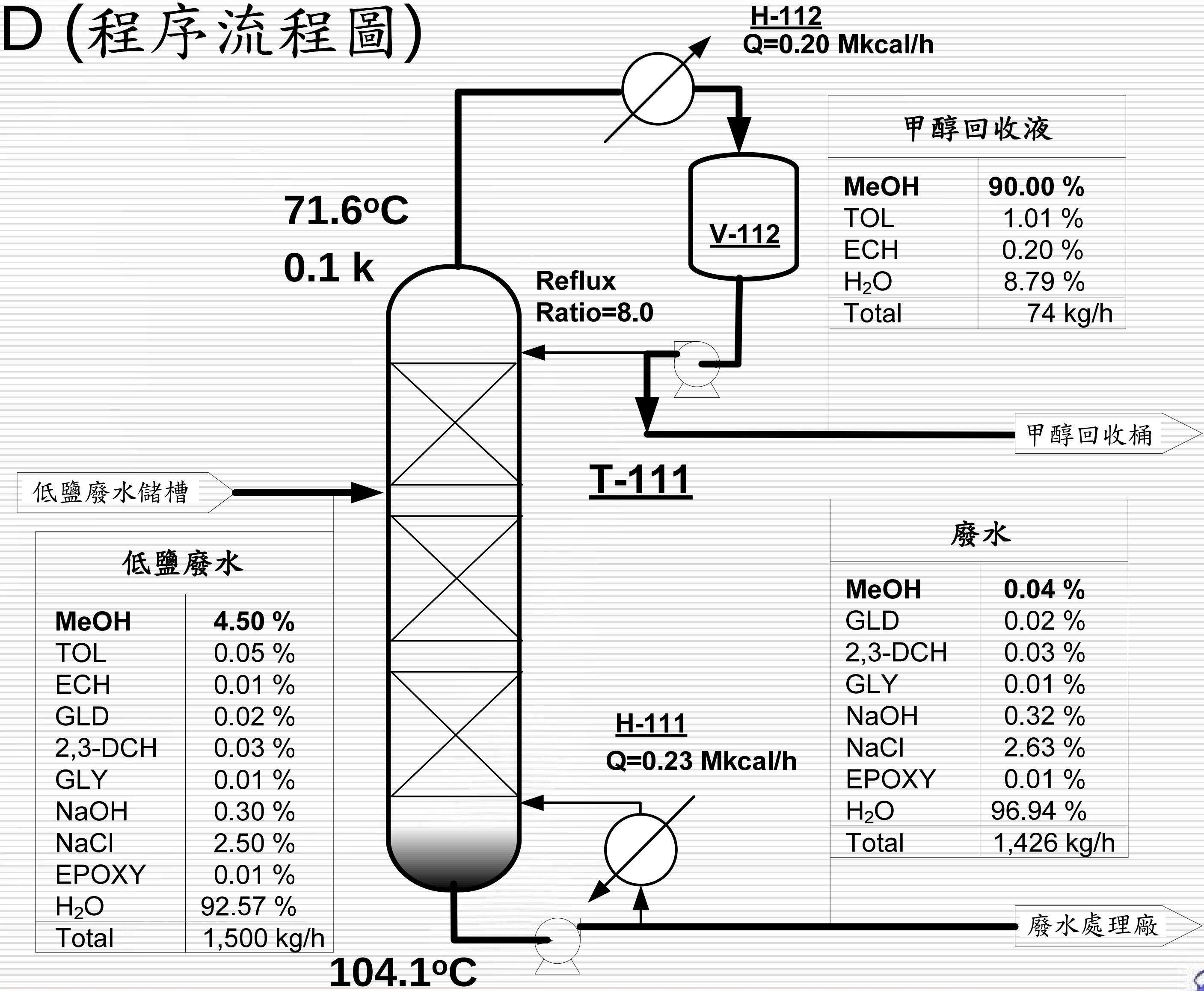
d. Piping & Instrument Diagram (管線及儀錶流程圖)

考慮實際操作 (包括開停車、正常運轉及緊急停車)、維修及工安、環保等需求下的所有相關設備、管線和儀錶

BFD (方塊流程圖)



PFD (程序流程圖)

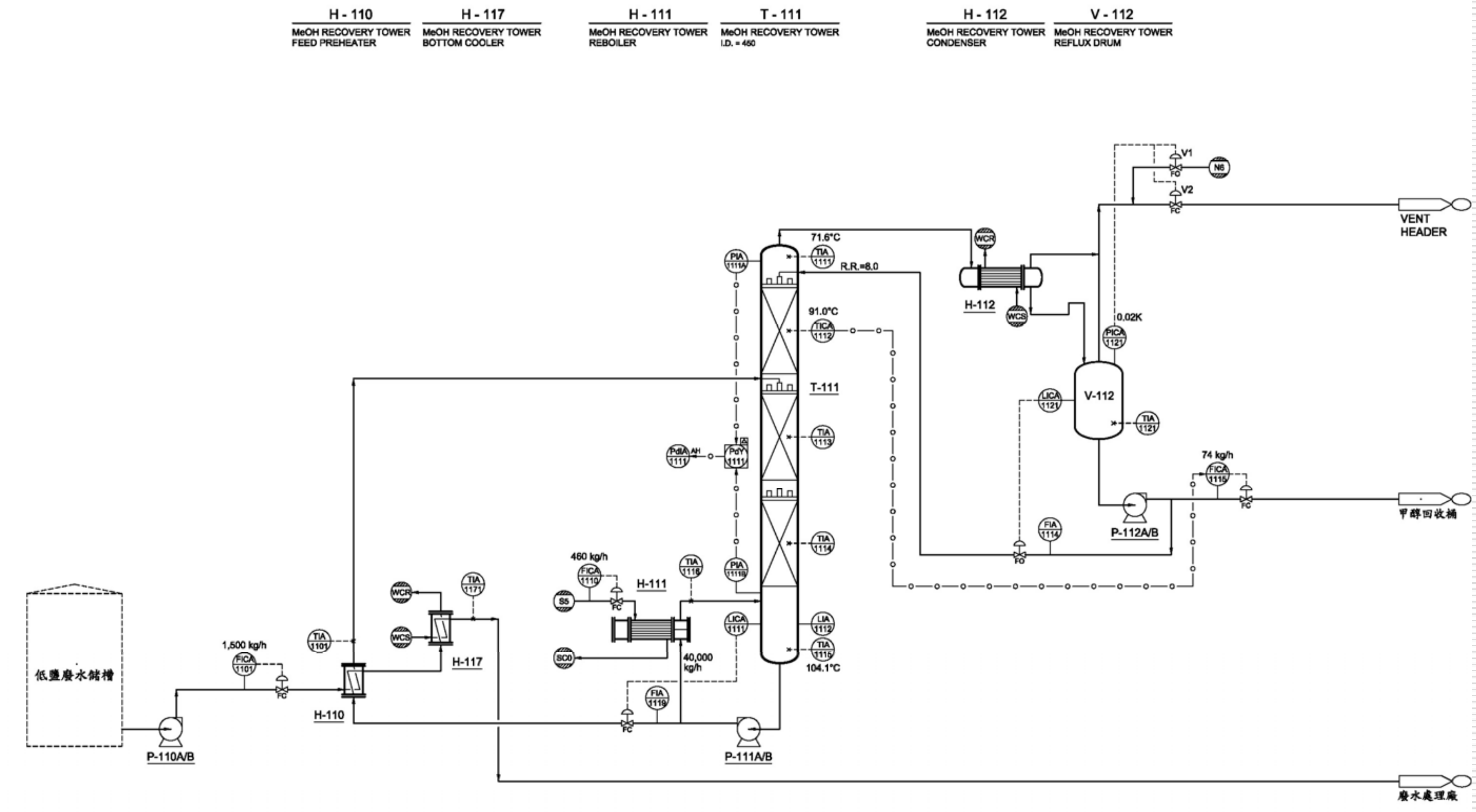


甲醇回收液	
MeOH	**90.00 %**
TOL	1.01 %
ECH	0.20 %
H_2O	8.79 %
Total	74 kg/h

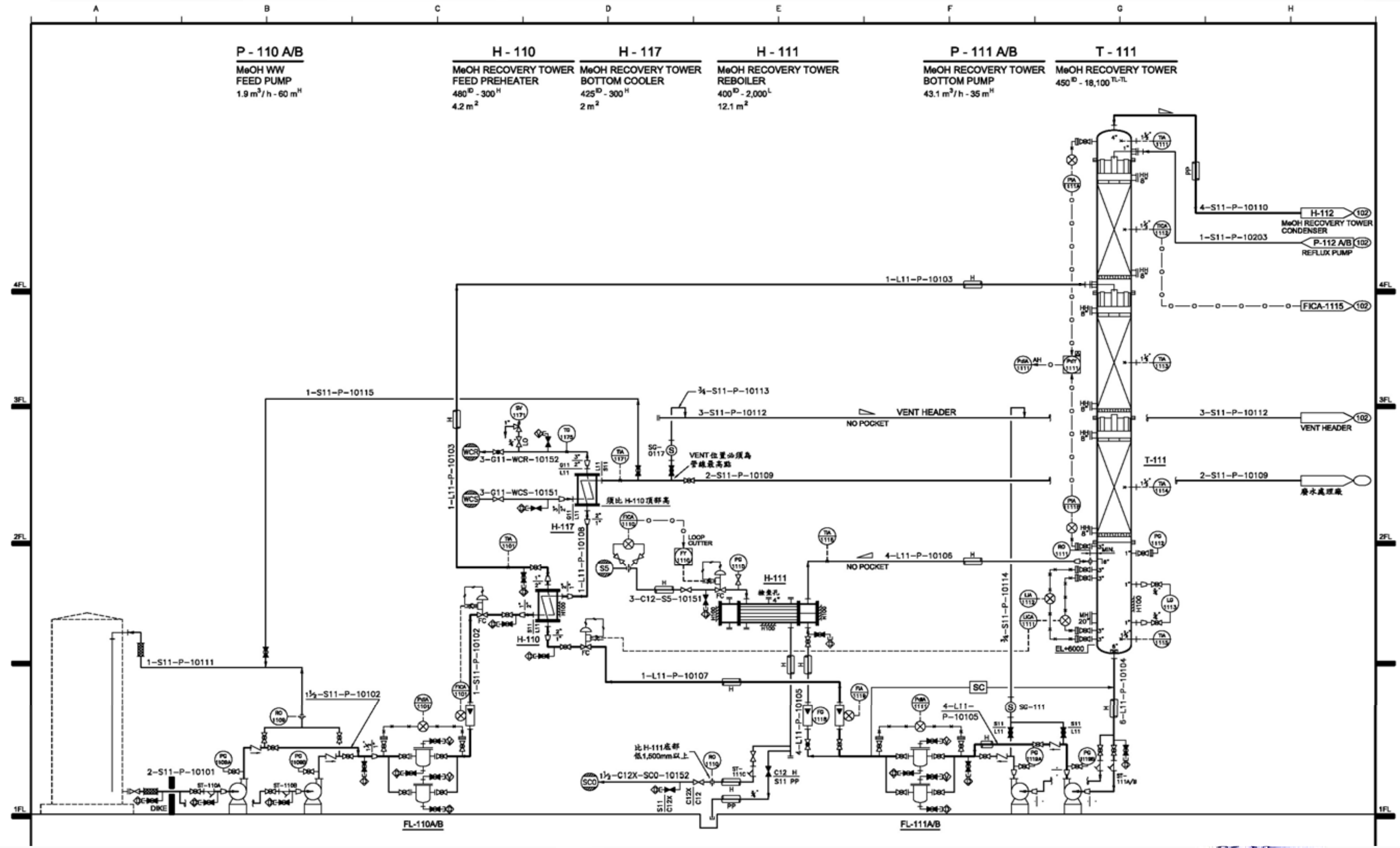
低鹽廢水	
MeOH	**4.50 %**
TOL	0.05 %
ECH	0.01 %
GLD	0.02 %
2,3-DCH	0.03 %
GLY	0.01 %
NaOH	0.30 %
NaCl	2.50 %
EPOXY	0.01 %
H_2O	92.57 %
Total	1,500 kg/h

廢水	
MeOH	**0.04 %**
GLD	0.02 %
2,3-DCH	0.03 %
GLY	0.01 %
NaOH	0.32 %
NaCl	2.63 %
EPOXY	0.01 %
H_2O	96.94 %
Total	1,426 kg/h

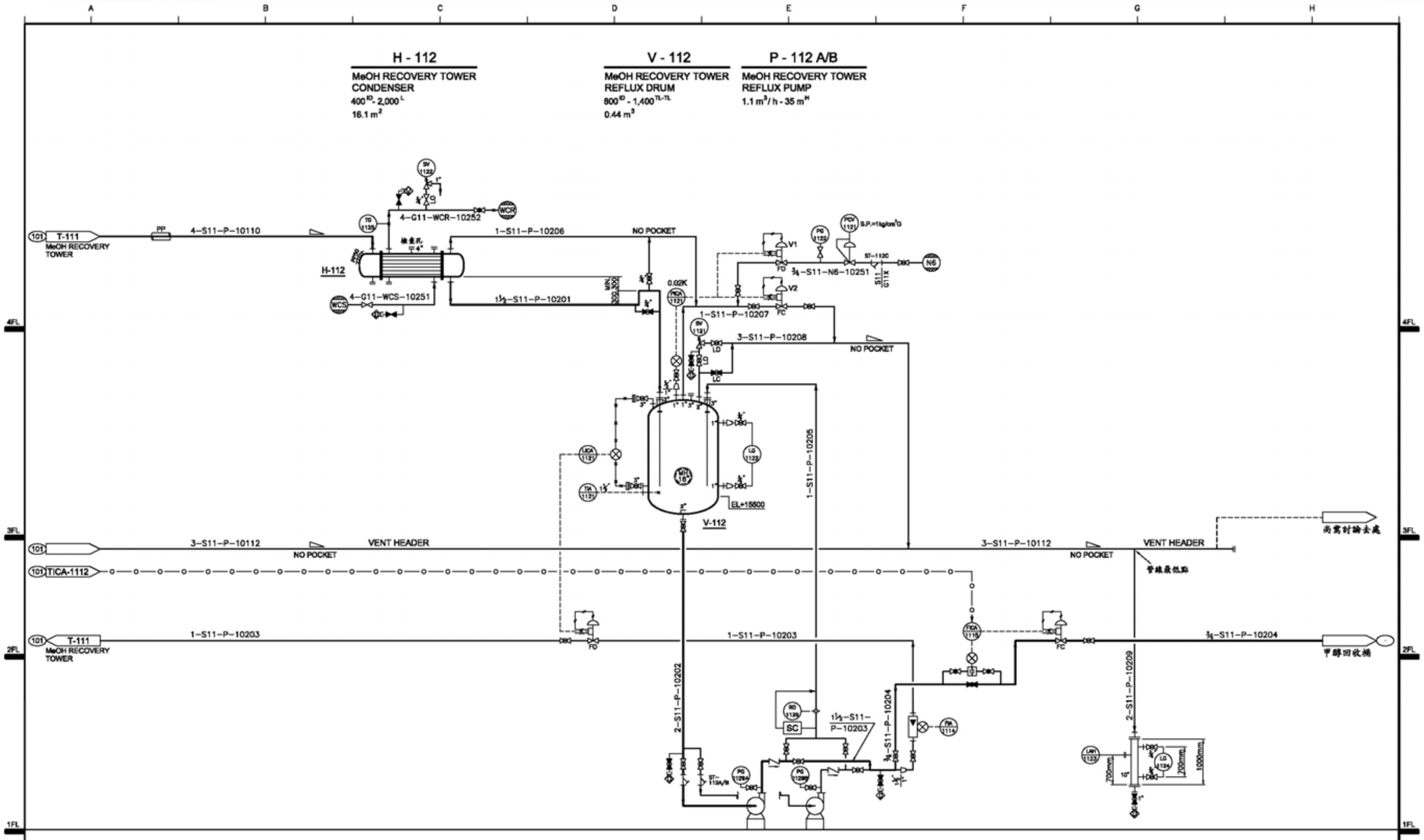
PCF (控制流程圖)


H - 110
MeOH RECOVERY TOWER FEED PREHEATER
H - 117
MeOH RECOVERY TOWER BOTTOM COOLER
H - 111
MeOH RECOVERY TOWER REBOILER
T - 111
MeOH RECOVERY TOWER I.D. = 450
H - 112
MeOH RECOVERY TOWER CONDENSER
V - 112
MeOH RECOVERY TOWER REFLUX DRUM
V1
N6
V2
VENT HEADER
71.6°C
R.R.=8.0
91.0°C
0.02K
T-111
H-112
V-112
74 kg/h
P-112A/B
甲醇回收桶
460 kg/h
H-111
40,000 kg/h
104.1°C
1,500 kg/h
H-117
H-110
低鹽廢水儲槽
P-110A/B
P-111A/B
廢水處理廠

P & ID (管線及儀錶流程圖)- Sheet 1

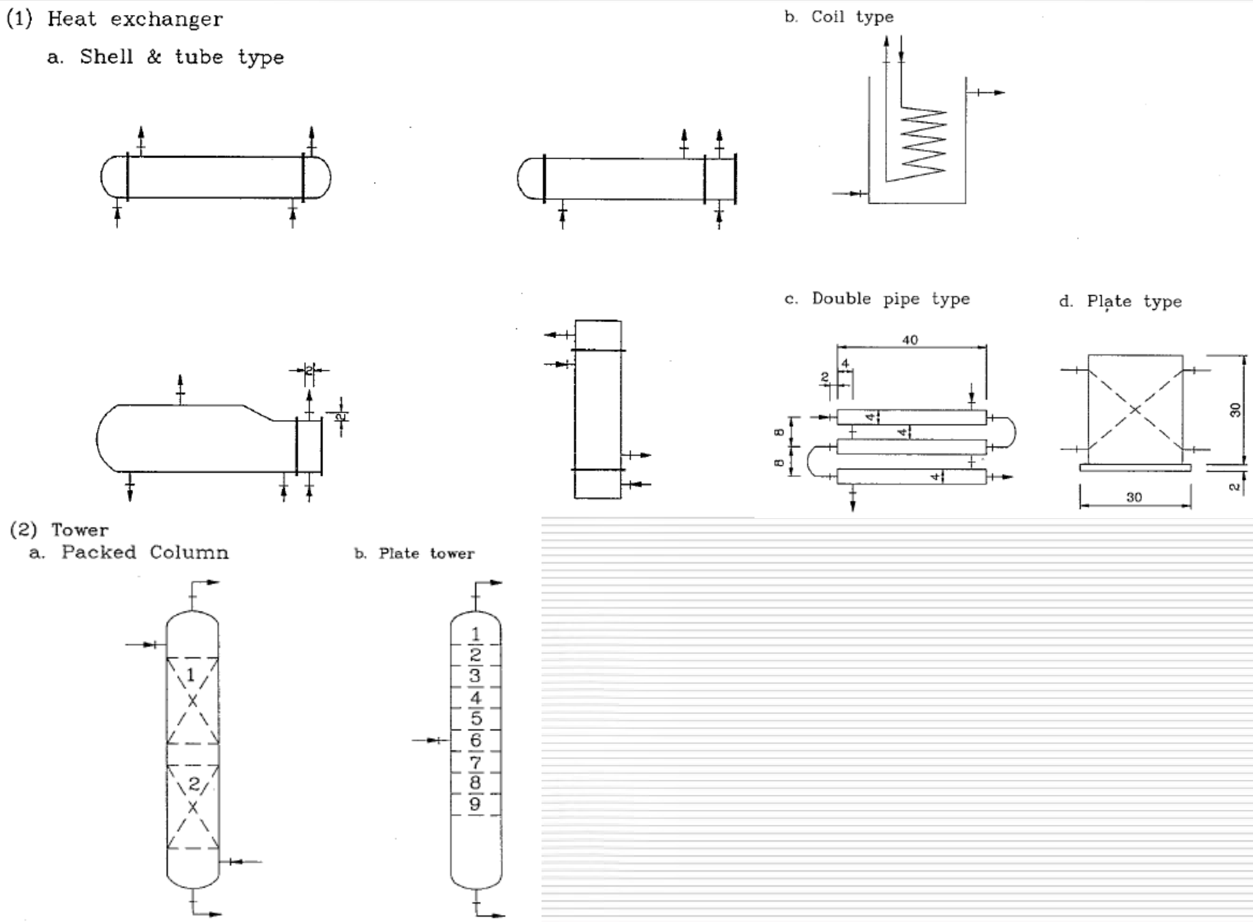


P & ID (管線及儀錶流程圖)- Sheet 2

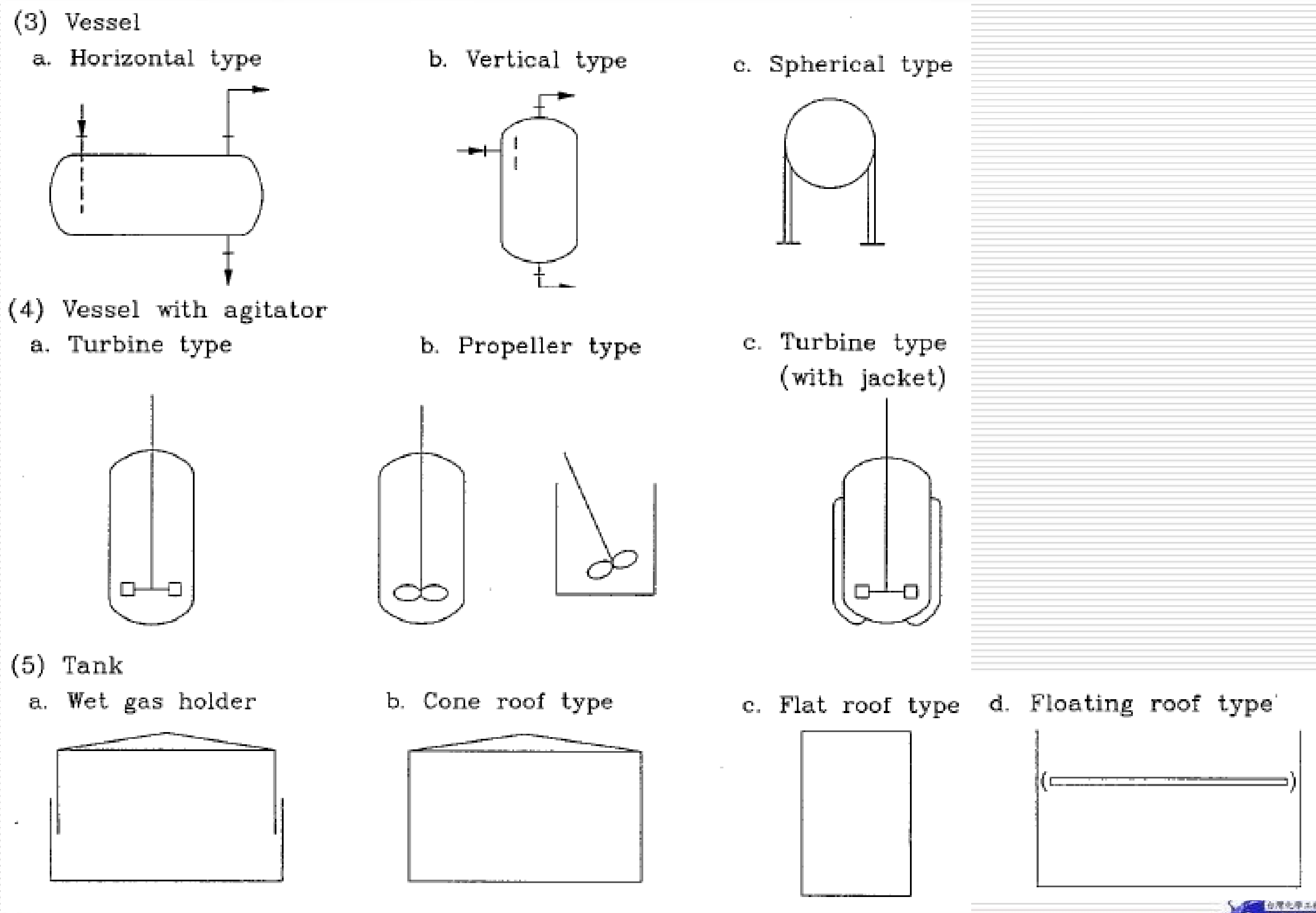


P & ID的內容主要分為設備、管線及儀錶三部份，常用的各種符號如下：

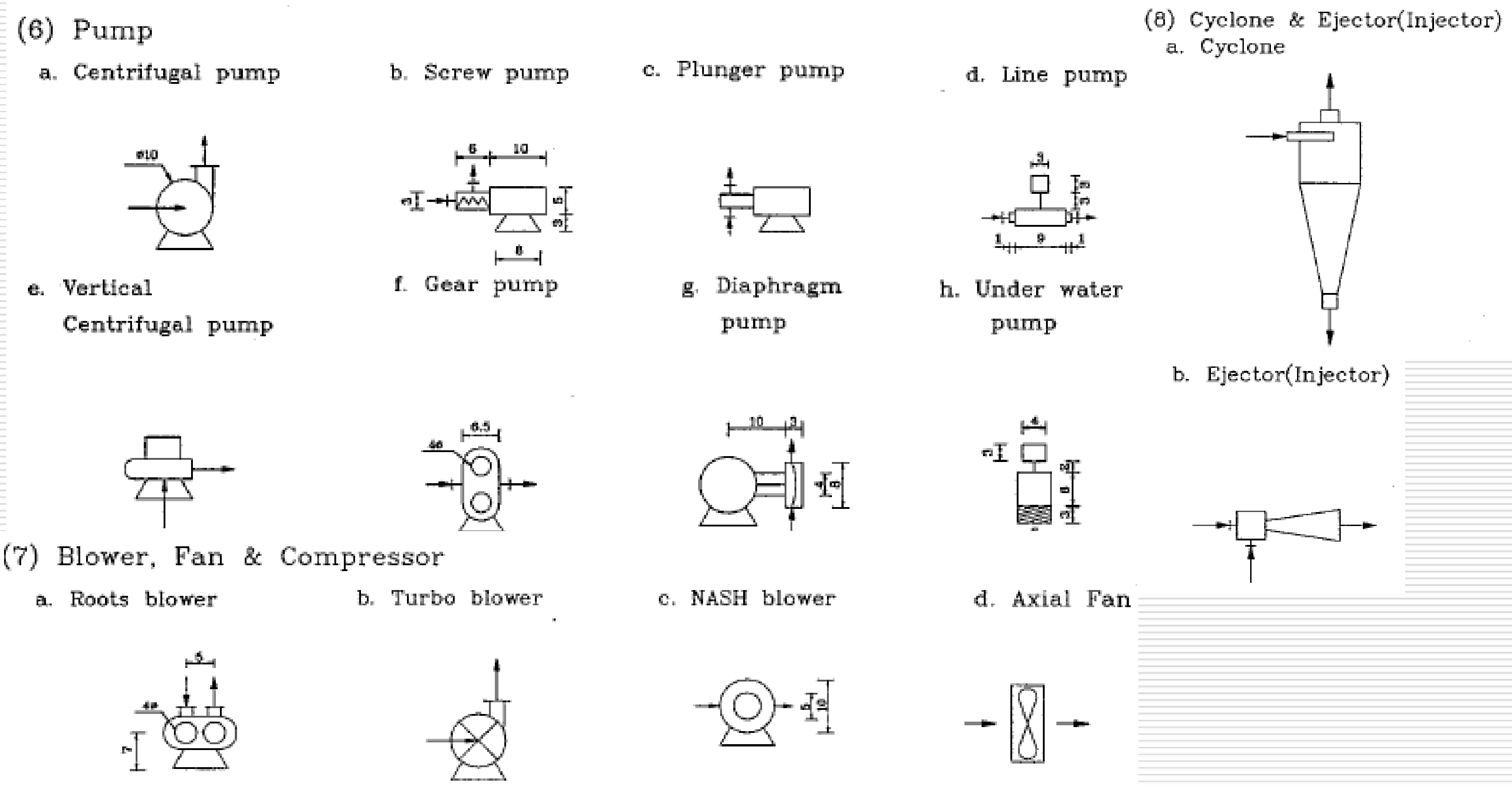
a. 設備部分



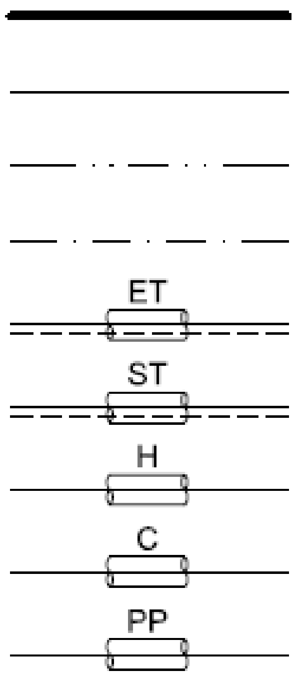
a. 設備部分 (續)


(3) Vessel
a. Horizontal type
b. Vertical type
c. Spherical type
(4) Vessel with agitator
a. Turbine type
b. Propeller type
c. Turbine type (with jacket)
(5) Tank
a. Wet gas holder
b. Cone roof type
c. Flat roof type
d. Floating roof type

a. 設備部分 (續)


(6) Pump
a. Centrifugal pump
b. Screw pump
c. Plunger pump
d. Line pump
e. Vertical Centrifugal pump
f. Gear pump
g. Diaphragm pump
h. Under water pump
(7) Blower, Fan & Compressor
a. Roots blower
b. Turbo blower
c. NASH blower
d. Axial Fan
(8) Cyclone & Ejector(Injector)
a. Cyclone
b. Ejector(Injector)

b. 管線部分



PRIMARY PIPING

SECONDARY PIPING

PIPING AND EQUIPMENT SHOWN ON OTHER P AND I DIAGRAMS

FUTURE PIPING AND EQUIPMENT

ELECTRIC TRACE AND INSULATE TO MAINTAIN TEMPERATURE AS INDICATED IN DEGREES CENTIGRADE

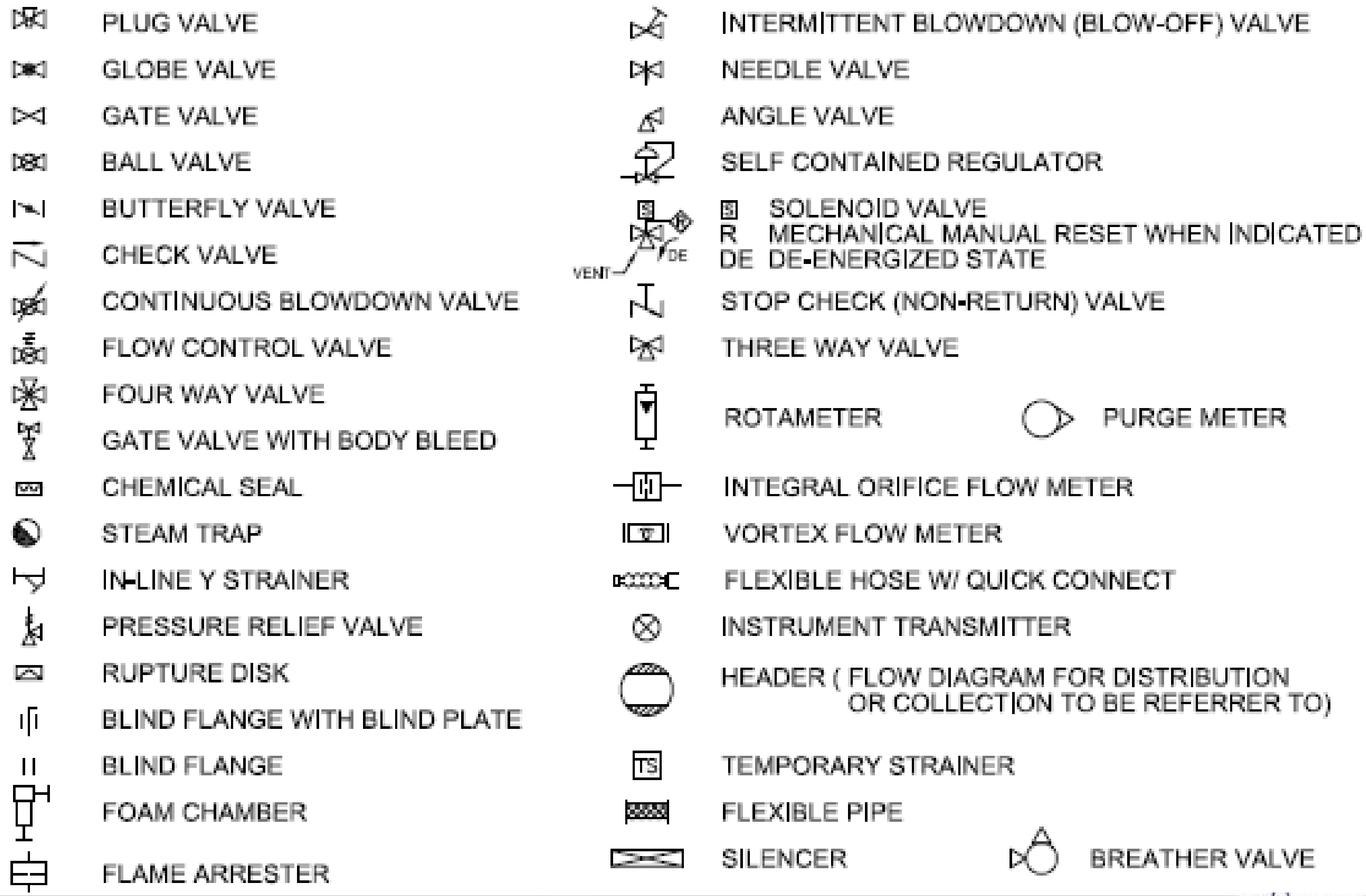
STEAM TRACE AND INSULATE

HOT INSULATE

COLD INSULATE

PERSONNEL PROTECTION

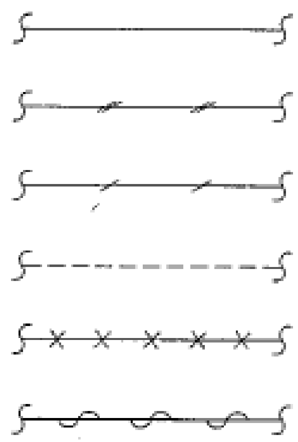
b. 管線部分 (續)


PLUG VALVE
GLOBE VALVE
GATE VALVE
BALL VALVE
BUTTERFLY VALVE
CHECK VALVE
CONTINUOUS BLOWDOWN VALVE
FLOW CONTROL VALVE
FOUR WAY VALVE
GATE VALVE WITH BODY BLEED
CHEMICAL SEAL
STEAM TRAP
IN-LINE Y STRAINER
PRESSURE RELIEF VALVE
RUPTURE DISK
BLIND FLANGE WITH BLIND PLATE
BLIND FLANGE
FOAM CHAMBER
FLAME ARRESTER
INTERMITTENT BLOWDOWN (BLOW-OFF) VALVE
NEEDLE VALVE
ANGLE VALVE
SELF CONTAINED REGULATOR
S SOLENOID VALVE
R MECHANICAL MANUAL RESET WHEN INDICATED
DE DE-ENERGIZED STATE
VENT
DE
STOP CHECK (NON-RETURN) VALVE
THREE WAY VALVE
ROTAMETER
PURGE METER
INTEGRAL ORIFICE FLOW METER
VORTEX FLOW METER
FLEXIBLE HOSE W/ QUICK CONNECT
INSTRUMENT TRANSMITTER
HEADER (FLOW DIAGRAM FOR DISTRIBUTION OR COLLECTION TO BE REFERRER TO)
TS
TEMPORARY STRAINER
FLEXIBLE PIPE
SILENCER
BREATHER VALVE

C. 儀錶部分

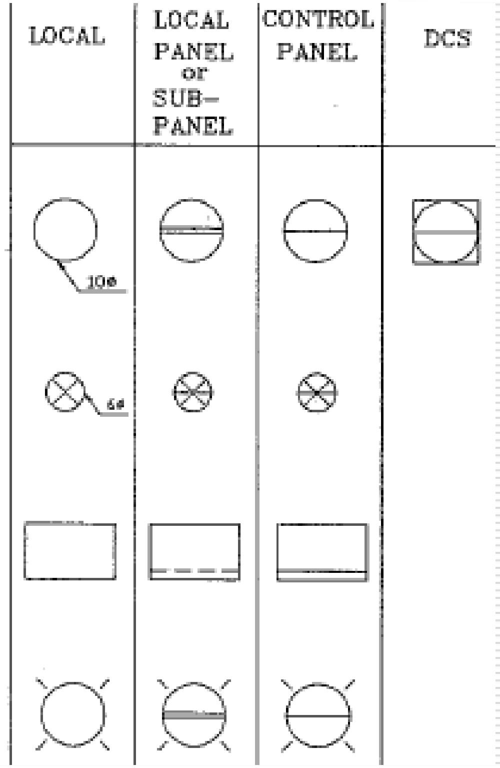
1) Instrument Line Symbol

- Lead or Impulse Piping
- Pneumatic Signal
- Hydraulic Signal
- Electrical Signal
- Capillary Tubing
- Radiation Signal

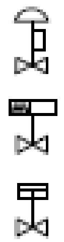


2) Balloon symbols

- Instrument
- Transmitter
- Computing Relay
- Pilot Lamp



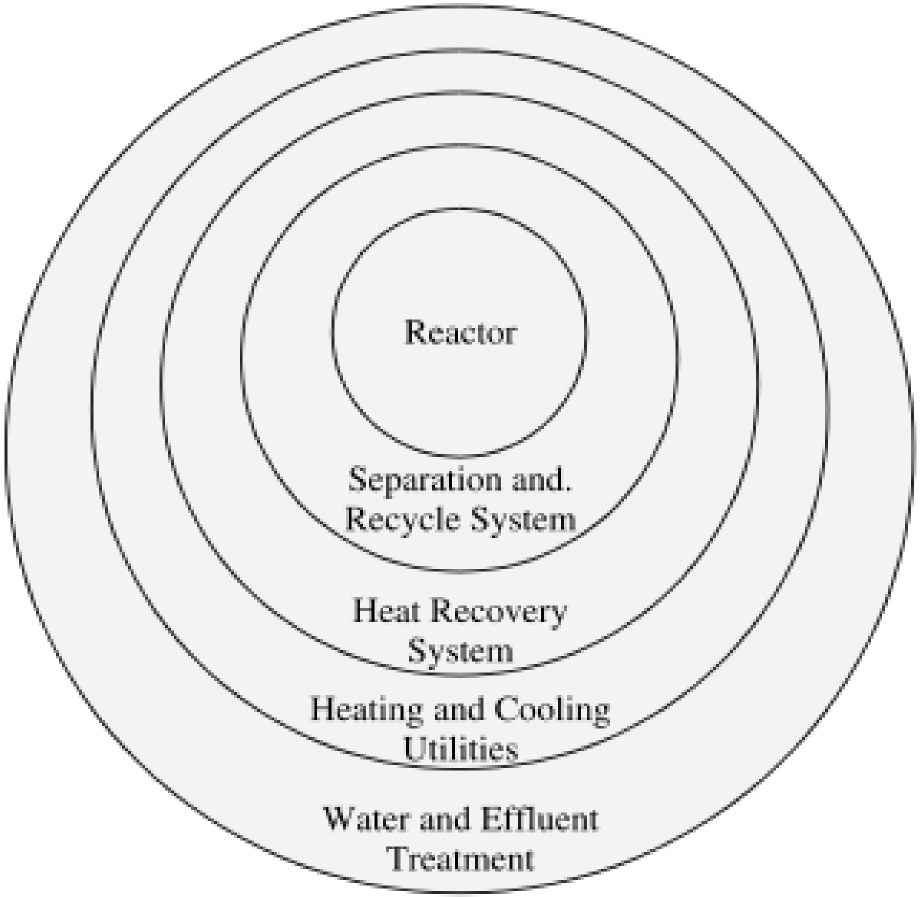
CONTROL VALVE SYMBOLS



- CONTROL VALVE W/ POSITIONER
- 90° CYLINDER ACTUATED SPRING LOADED
- AXIAL CYLINDER ACTUATED

三、程序設計概念

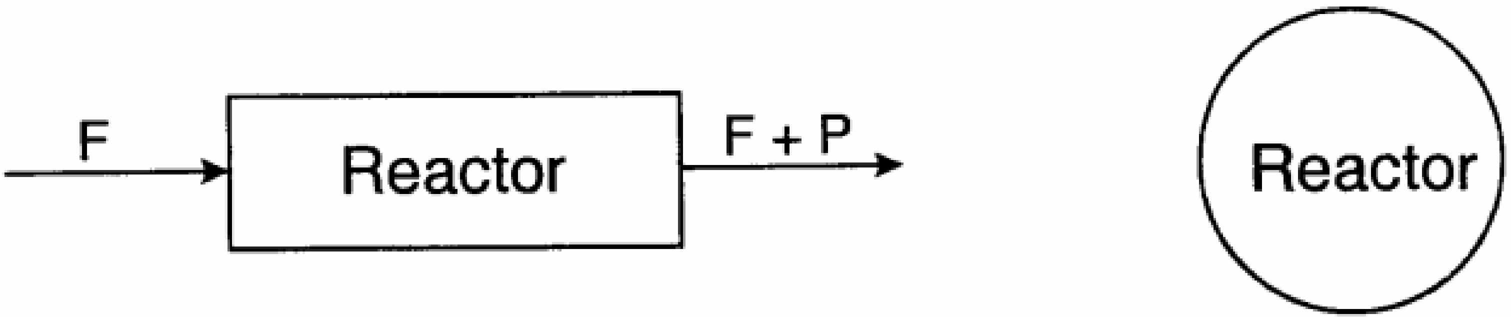
- Onion Model



A reactor is needed before the separation and recycle system can be designed and so on

- Onion Model

1st Layer : Reactor

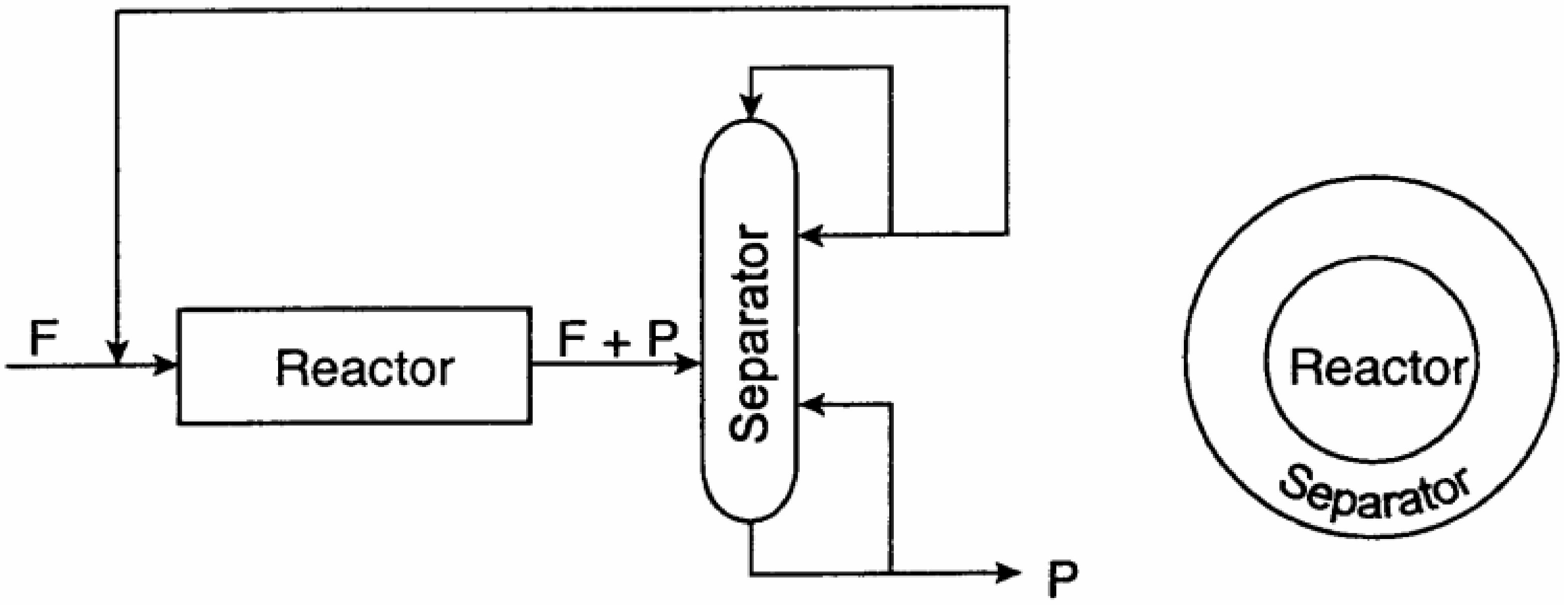


Process design starts with the reactor

反應器無法模擬，須由 **Bench** 或 **Pilot** 實驗得知

• Onion Model

2nd Layer : Separation and recycle system

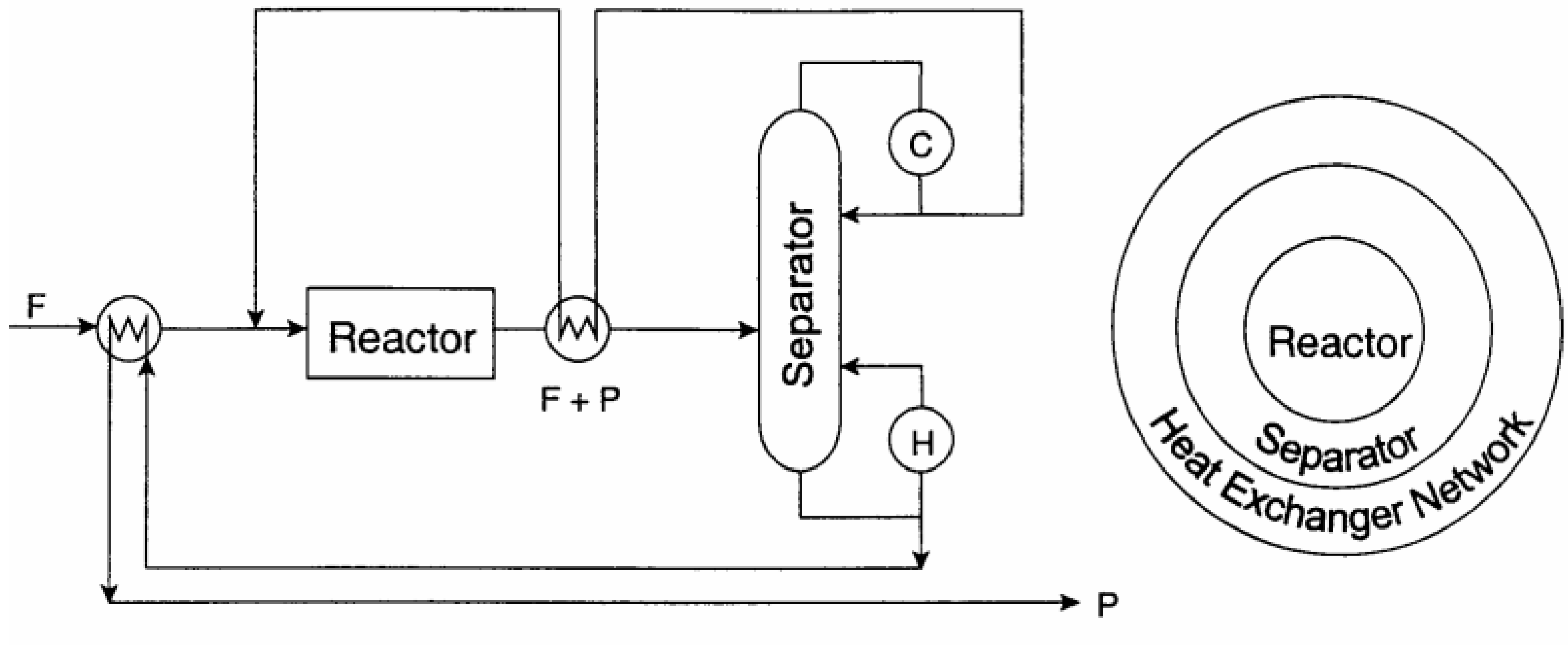


Once we know the reactor design,
we need to specify the separation system

此步驟包含探討製程最佳化

- Onion Model

3rd Layer: Heat recovery system

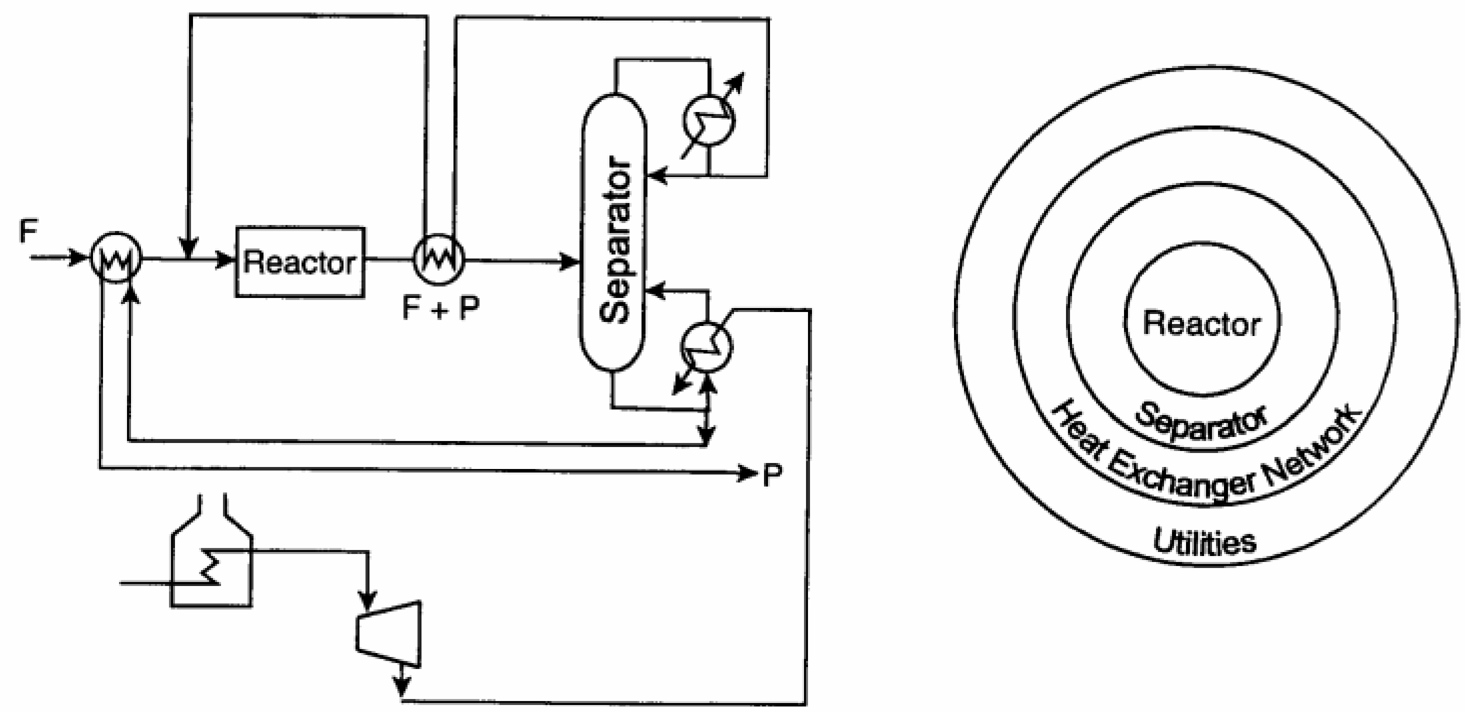


Once we know reactor and separator design, we know material and energy balance ➪ Heat Exchanger Network

此步驟須考慮初開車及停車情況

• Onion Model

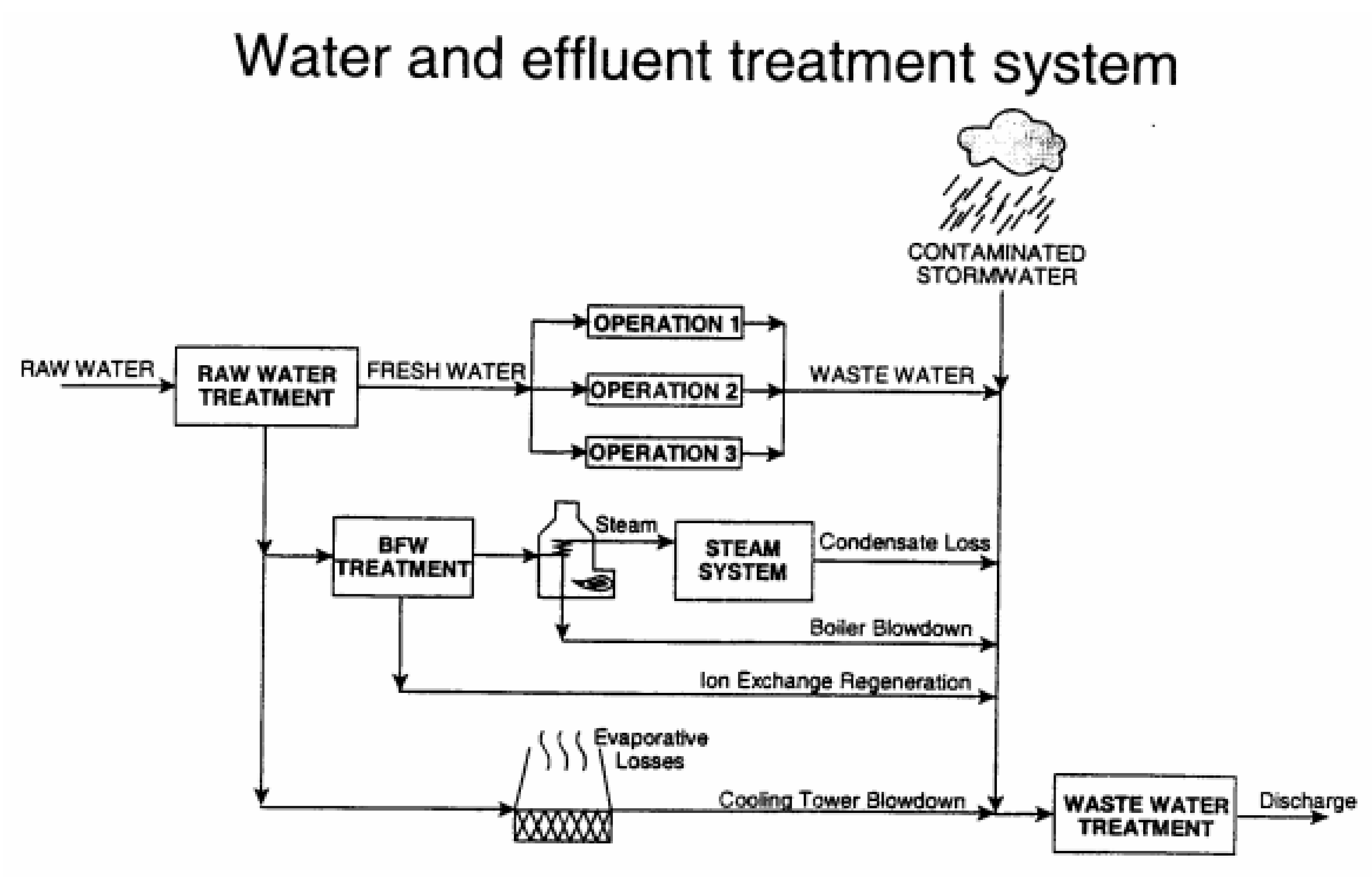
4th Layer: Heating and cooling utilities



Once we have the process and HEN design
➪ Utilities

- Onion Model

5th Layer: Water and effluent treatment



四、程序最適化

- 製程最適化

  反應：提高轉化率、減少副產物

  分離：連續或批次、新型分離技術、設備型式選擇、熱整合、最佳控制策略

- 設備最適化

  基本設計分析

  設備計算最適化

製程最適化 - Hierarchical structure

Hierarchy of decisions

1. Batch versus continuous
2. Input-output structure of the flowsheet

Reactor

3. Recycle structure of the flowsheet
4. General structure of the separation system
   a. Vapor recover system
   b. Liquid recovery system

Separation & Recycle System

5. Heat-exchanger network

Heat Recovery System

Advantages: Estimate the cost as proceed through the levels in the hierarchy

Objective: Find an optimum design through a systematic analysis

1. Batch vs Continuous

Distinction between batch and continuous process is "Fuzzy"

(a) Production rate
- continuous process > 10^7 lb/year
- batch process < 10^6 lb/year

(b) Market forces
(c) Operation problems
(d) Multiple operations in a single equipment

2. Input Output Structure

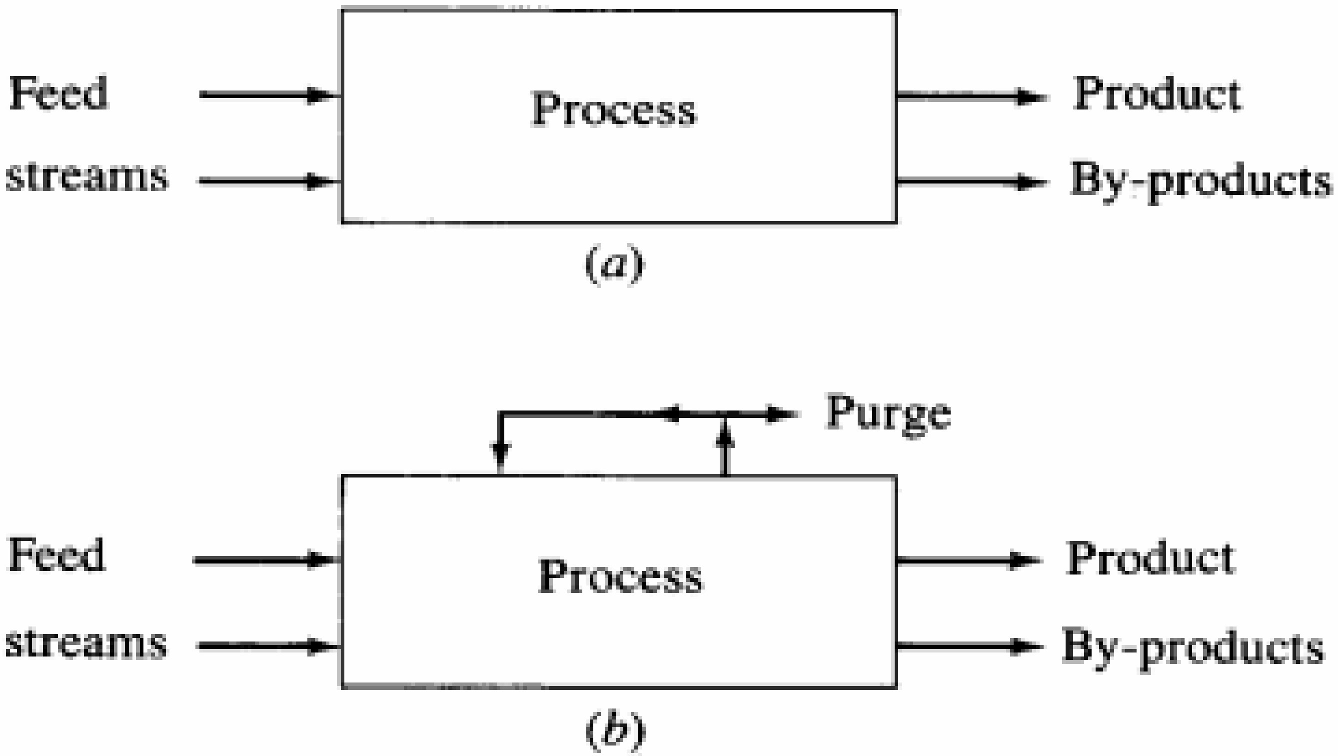


FIGURE 5.1-1
Input-output structure of flowsheet. [*From J. M. Douglas, AIChE J.*, **31**: *353 (1985)*.]

Some design guides for purification of feeds :

a. If feed impurity is

- not inert and is present in large amounts
- catalyst poison

→ **remove it**

b. If feed impurity is

- present in a gas feed
- a product or by product components
- form a azeotrope with reactant
- easier to separate from the product than the feed

→ **it is better to feed the process**

製程最適化 - 範例 1

The reactions to produce benzene from toluene are

$$\text{Toluene} + H_2 \rightarrow \text{Benzene} + CH_4$$

$$2\text{Benzene} \rightleftharpoons \text{Diphenyl} + H_2$$

Recycling the reversible by-product Diphenyl must oversize all the equipment in the recycle loop, however, removing it would increase raw material cost of toluene.

☆ ***The decision involves an economic trade-off between raw material losses to by-product diphenyl and increased recycle costs. No simple design guideline is available .***

3. Recycle Structure

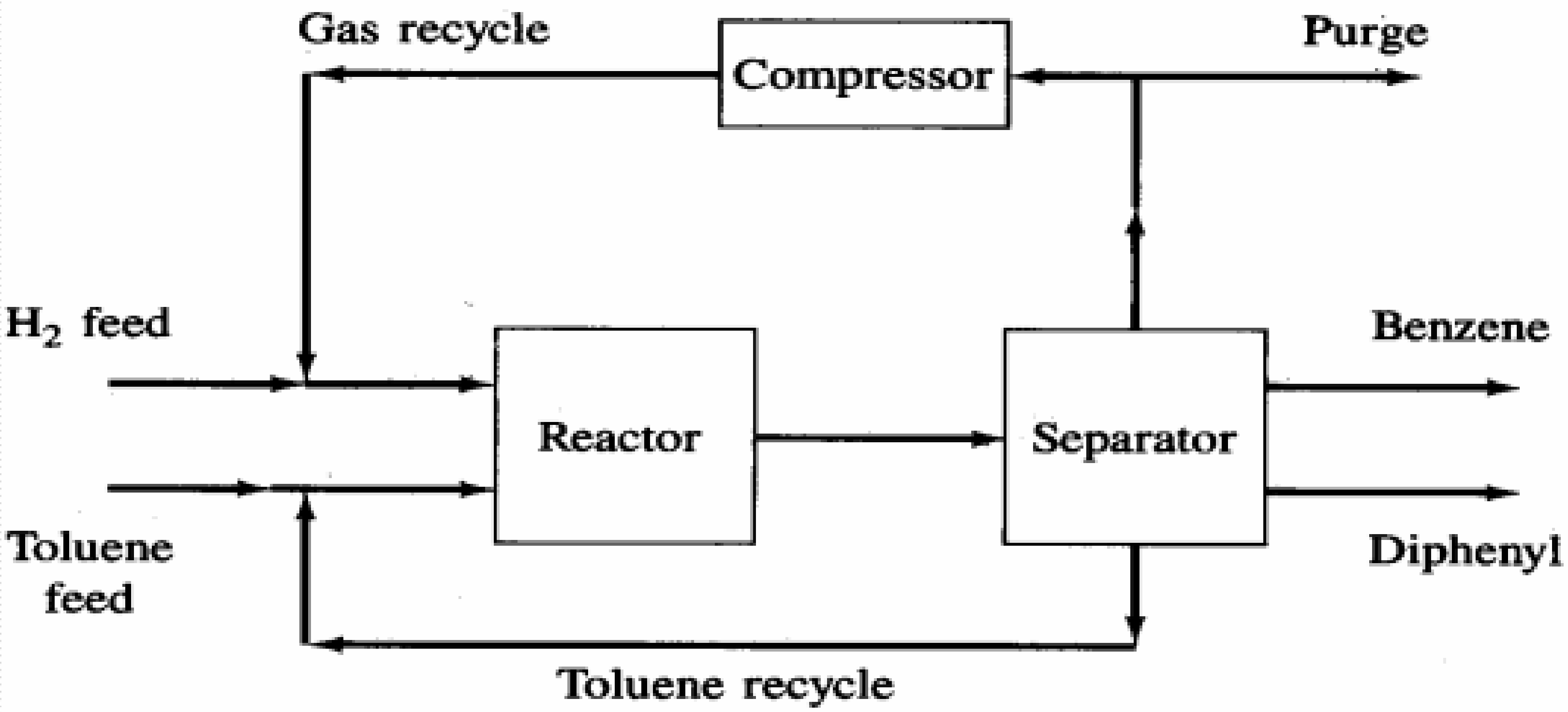


FIGURE 6.1-2
HDA recycle structure. [*From J. M. Douglas, AIChE J.*, **31**: *353 (1985)*.]

製程最適化 - 範例 2

What excess quantity of isobutane is optimum at the reactor inlet ?

Reactions are listed as below:

Butene + Isobutane → Isooctane

Butene + Isooctane → C_{12}

Use excess isobutane leads to an improved selectivity to produce isooctane, but the larger cost to recover and recycle isobutane.

☆ ***An optimum amount of excess must be determined from an economic analysis***

4. Separation System

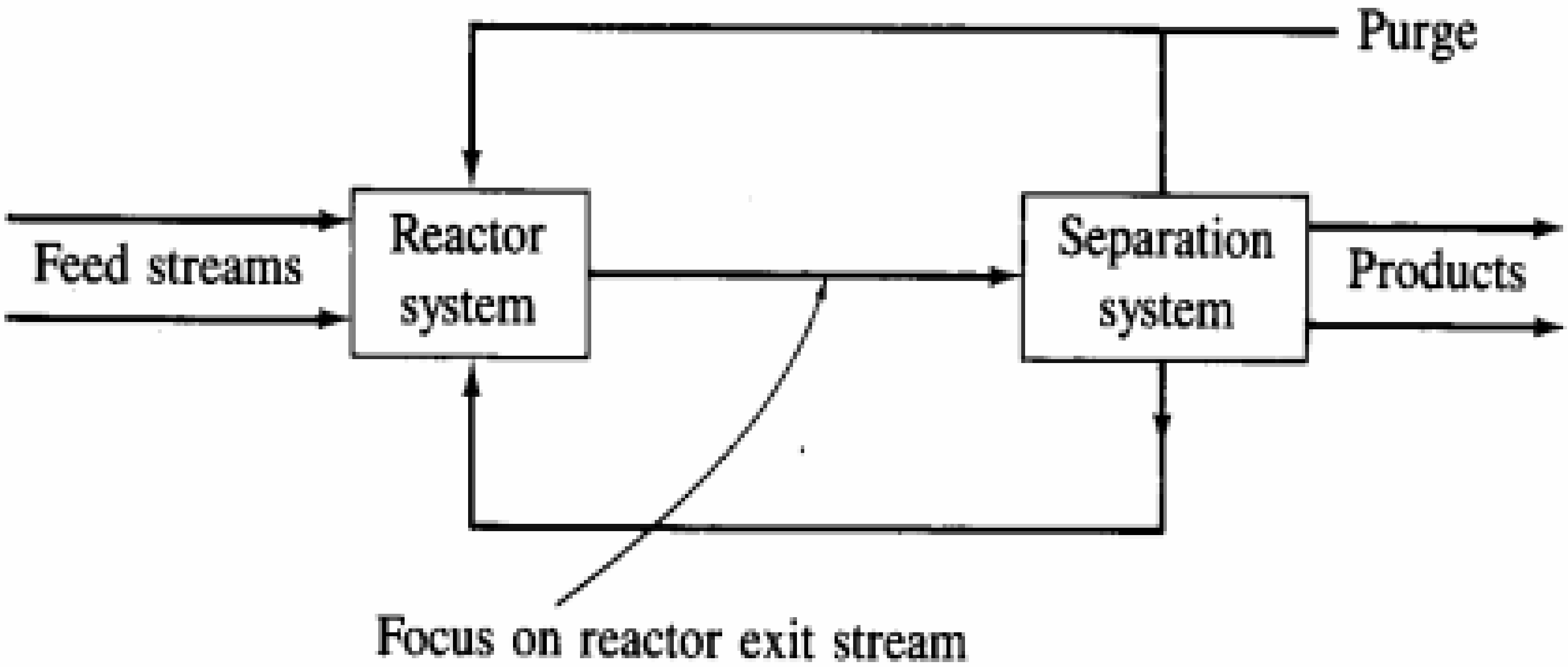


FIGURE 7.1-1
Phase of the reactor effluent stream.

Case 1 : reactor exit stream is liquid
Case 2 : reactor exit stream is vapor & liquid
Case 3 : reactor exit stream is vapor

Case 1: Reactor exit is liquid

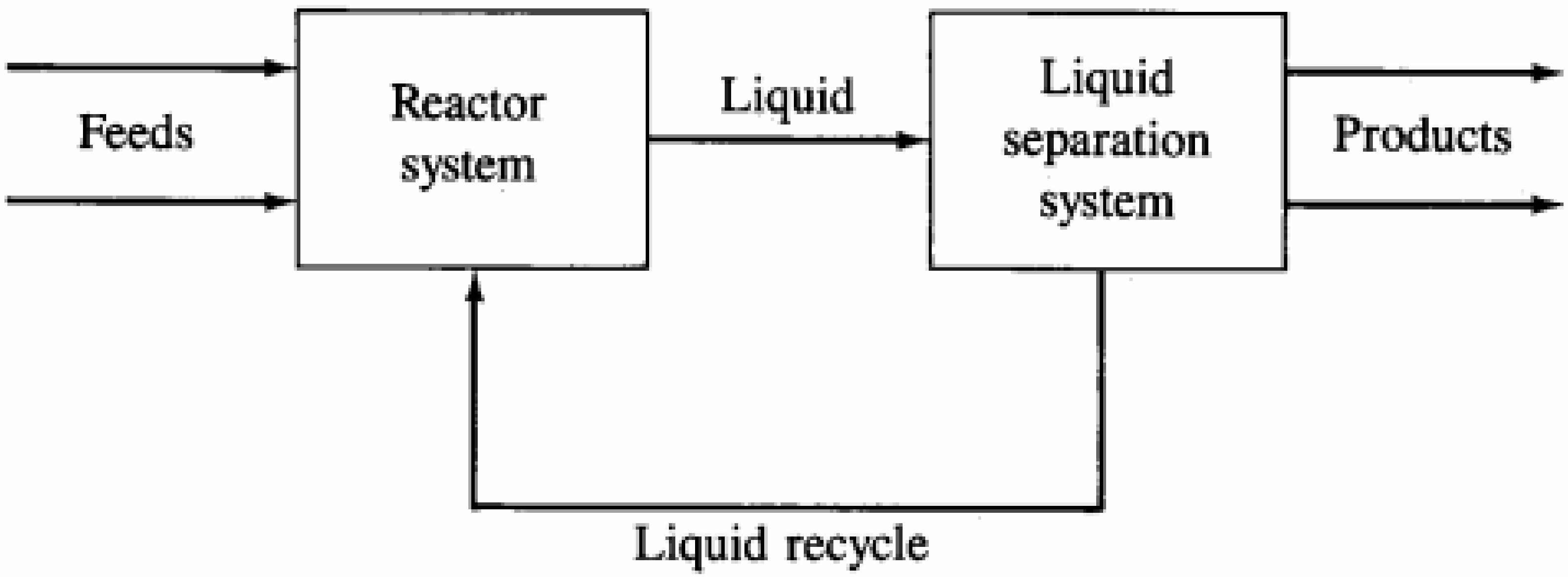


FIGURE 7.1-2
Reactor exit is liquid. [*From J. M. Douglas, AIChE J.*, **31**: *353 (1985)*.]

Liquid Separation System: distillation column, extraction units, azeotropic distillation

Case 2: Reactor exit is vapor and liquid

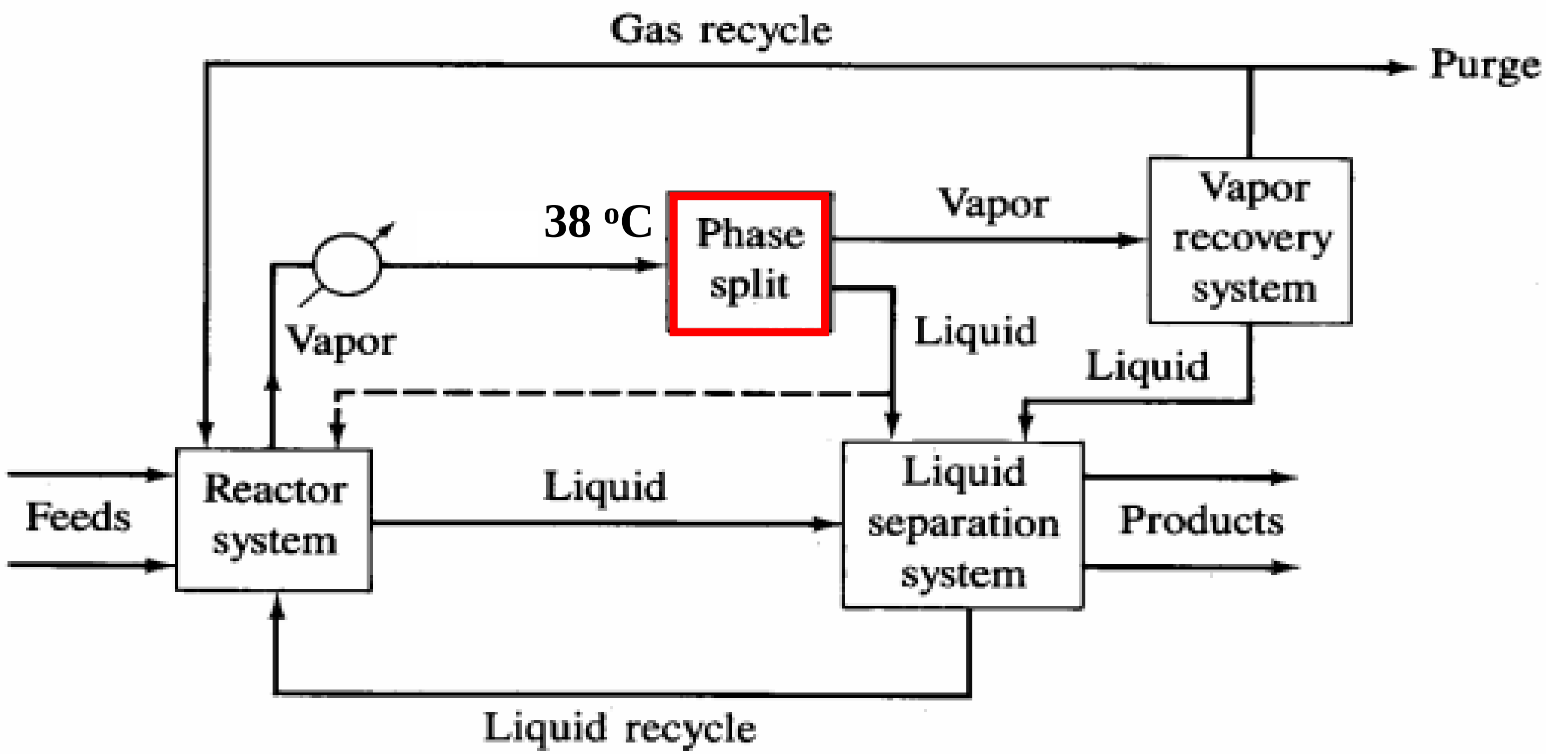


FIGURE 7.1-3
Reactor exit is vapor and liquid. [*From J. M. Douglas, AIChE J.*, **31**: *353 (1985)*.]

Use reactor as a phase splitter and cool reactor vapor to 38 ºC

Case 3: Reactor exit is vapor

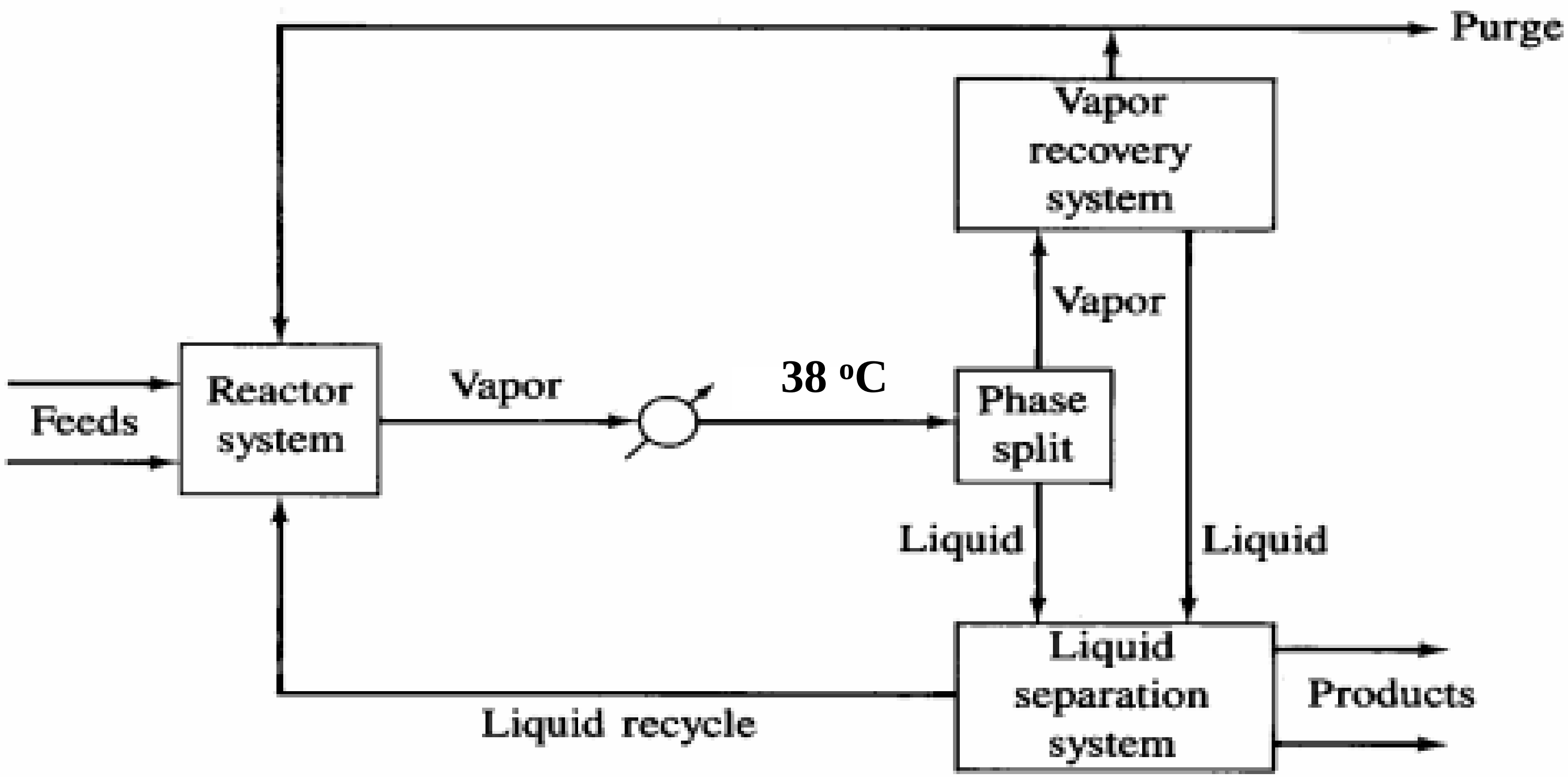


FIGURE 7.1-4
Reactor exit is vapor. [*From J. M. Douglas, AIChE J.*, **31**: *353 (1985).*]

Phase Splits are the cheapest method of separation

Vapor Recovery System

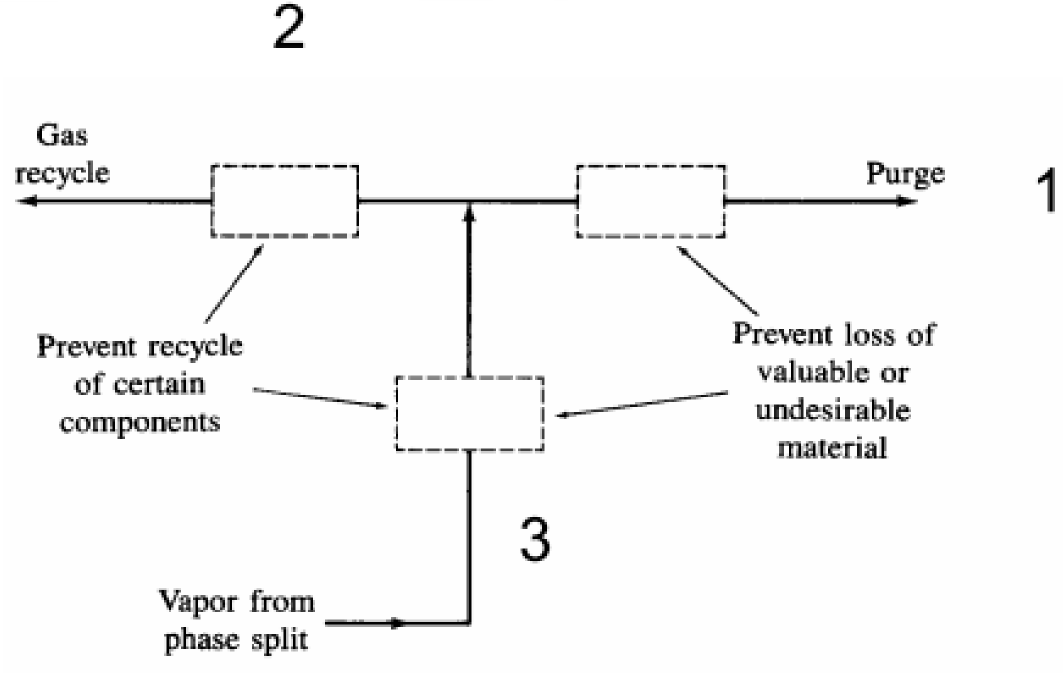


FIGURE 7.2-1
Vapor recovery system location.

Type: condensation
absorption
adsorption
membrane
reaction system
(removing CO_2, H_2S)

Vapor Recovery System (cont'd)

Location 1: If significant amount of valuable materials are being lost in the purge.
Location 2: If materials are deleterious to the reactor operation.
Location 3: Both factors are valid.
None: Both factors do not exist.

Liquid Separation System

Why use distillation?

- ability to handle a wide range of throughput
- ability to handle a wide range of feed concentrations
- ability to produce high product purity

Separations not suited to distillation:

- low molecular weight materials
- high molecular weight heat sensitive materials
- separation of components with a low concentration
- separation of classes of components
- relative volatility <1.1

$$\alpha = \frac{K_i}{K_j} = \frac{y_i / x_i}{y_j / x_j} = \frac{\gamma_i P_i^{sat}}{\gamma_j P_j^{sat}}$$ @ moderate pressure

Liquid Separation System (cont'd)

- **Light ends**

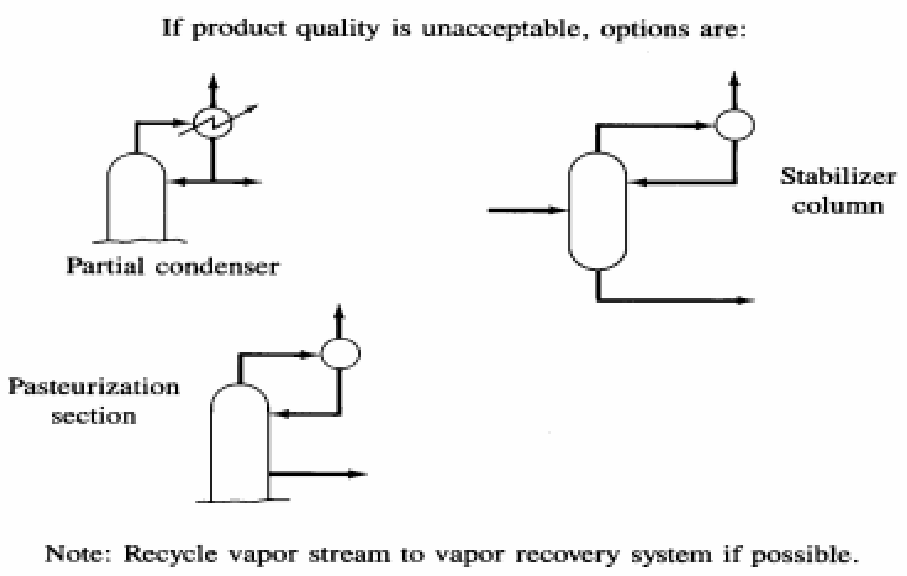


FIGURE 7.3-1
Alternatives for removing light ends.

Method

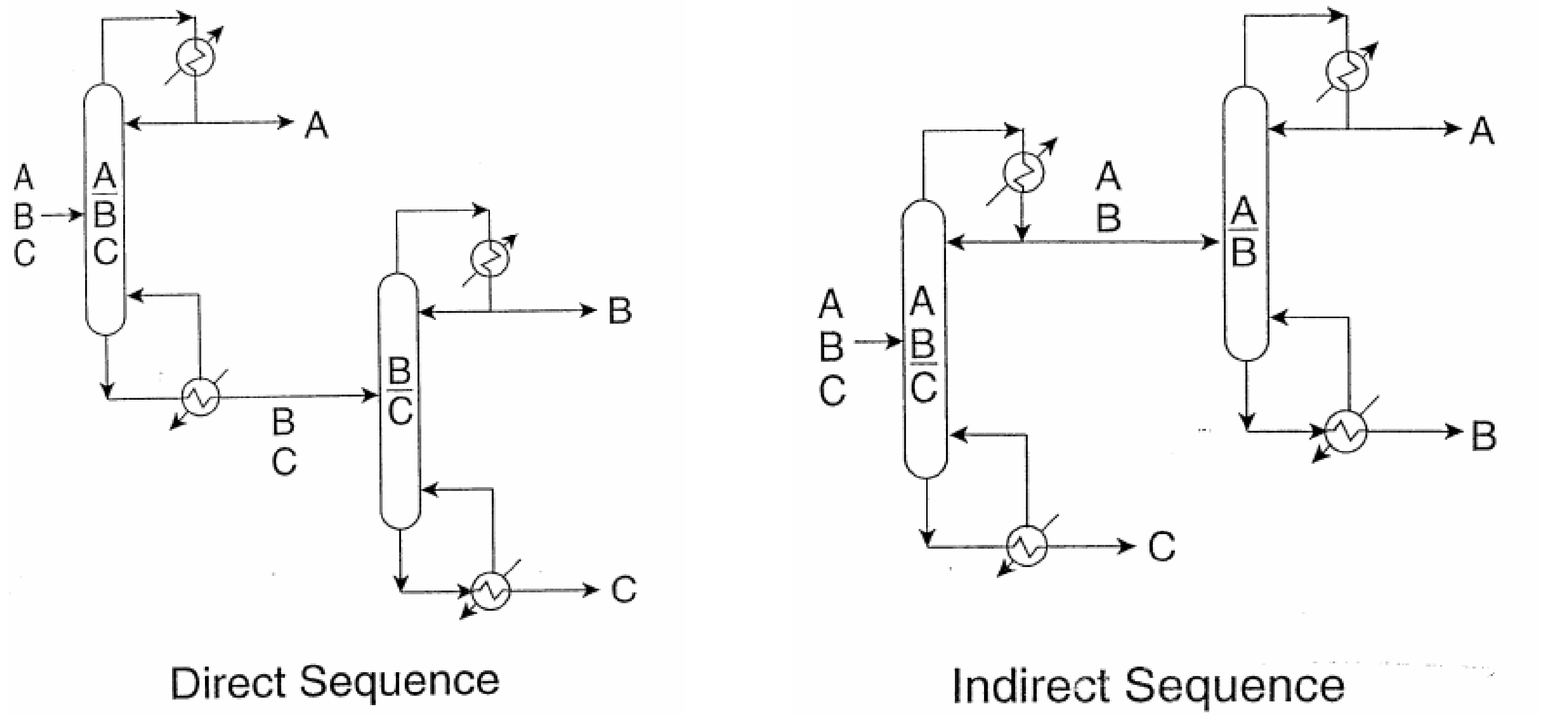
- drop the pressure for removal
- partial condenser
- pasteurization column
- stabilizer column

Destination

- vent
- flare
- vapor recovery system

- General heuristic for column sequencing
  - Remove corrosive component as soon as possible
  - Remove reactive components or monomers as soon as possible
  - Remove products as distillates
  - Remove recycle streams as distillates, particularly if they are recycled to a packed reactor

• 製程最適化 - Distillation sequencing



the lightest component is taken overhead in each column

the heaviest component as bottom product in each column

- Heuristics for column sequencing
  - Most plentiful first
  - Lightest first
  - High-recovery separations last
  - Difficult separations last
  - Favor equimolar splits
  - Next separation should be cheapest

製程最適化 - 範例 3

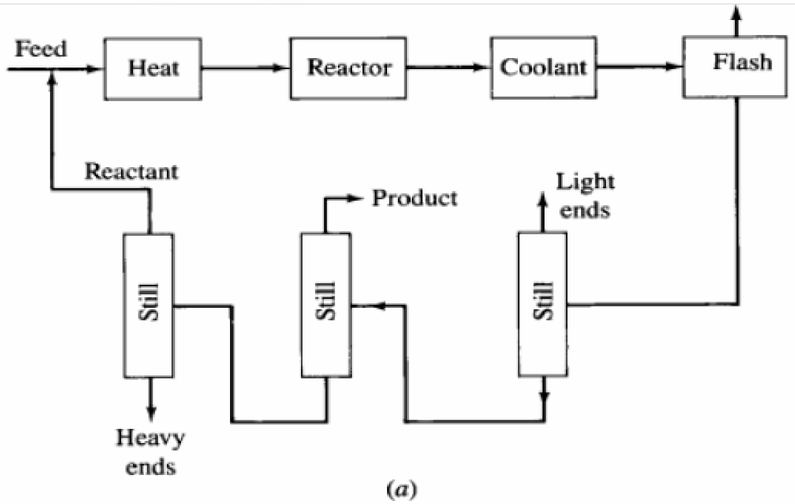
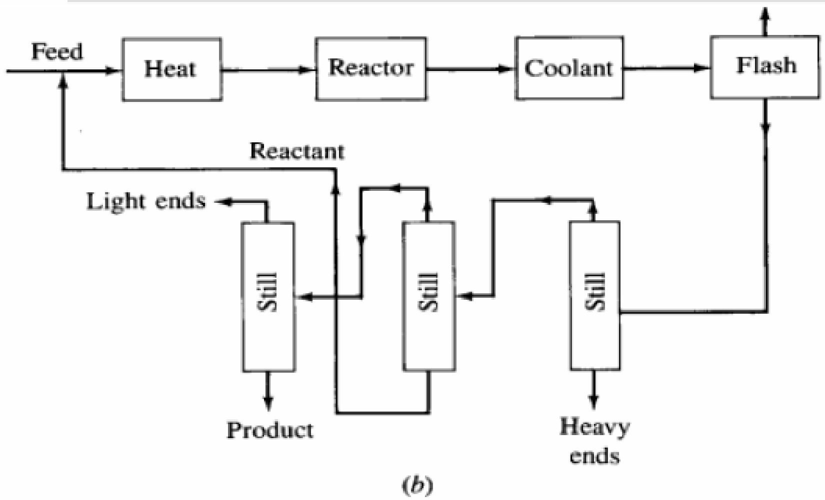


FIGURE 7.3-4
Sequence selection changes recycle costs.

製程流體沸點高低：
Light ends < Product < Reactant < Heavy ends

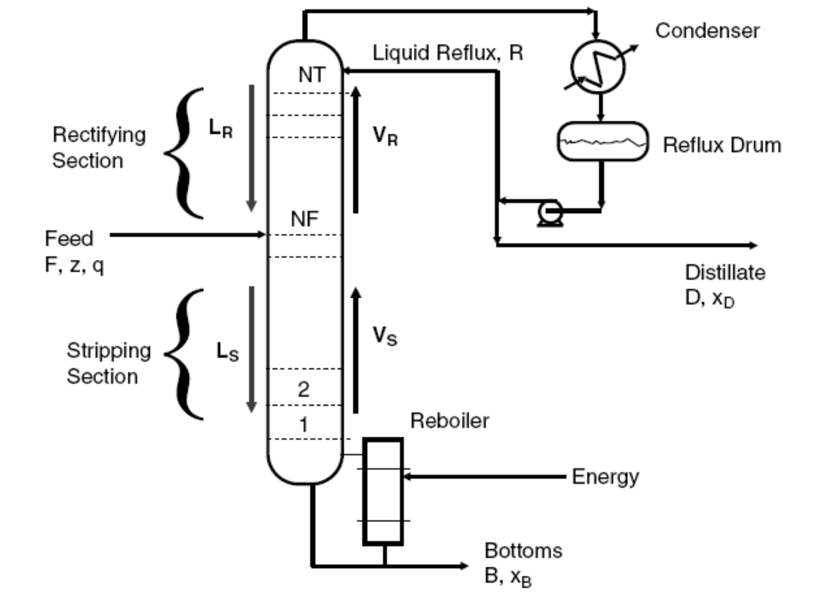
☆ ***Select the sequence that minimizes the number of columns in a recycle loop***



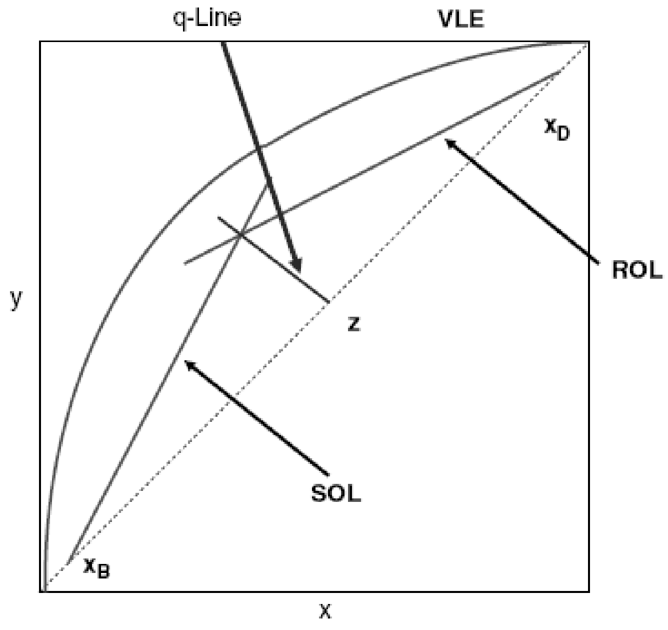
- 設備最適化 – 蒸餾塔

兩成分蒸餾塔理論板數計算

McCabe Thiele method assume Constant molar overflow

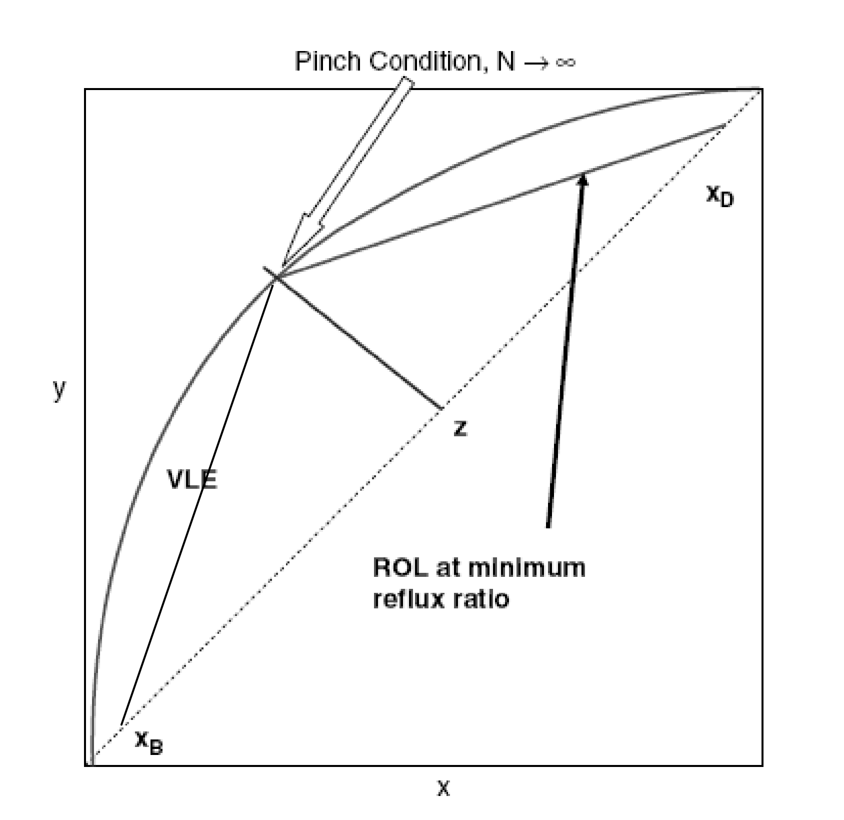


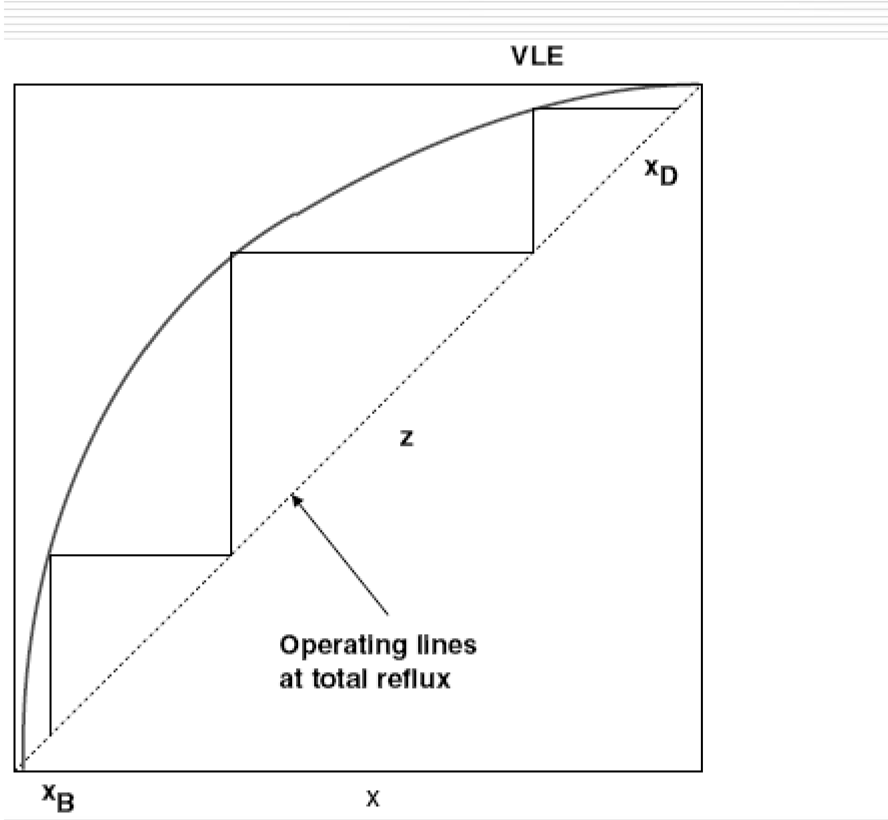
$$\text{Slope ROL} = \frac{L_R}{V_R} = \frac{R}{R+D} = \frac{RR}{1+RR}$$



McCabe Thiele Method

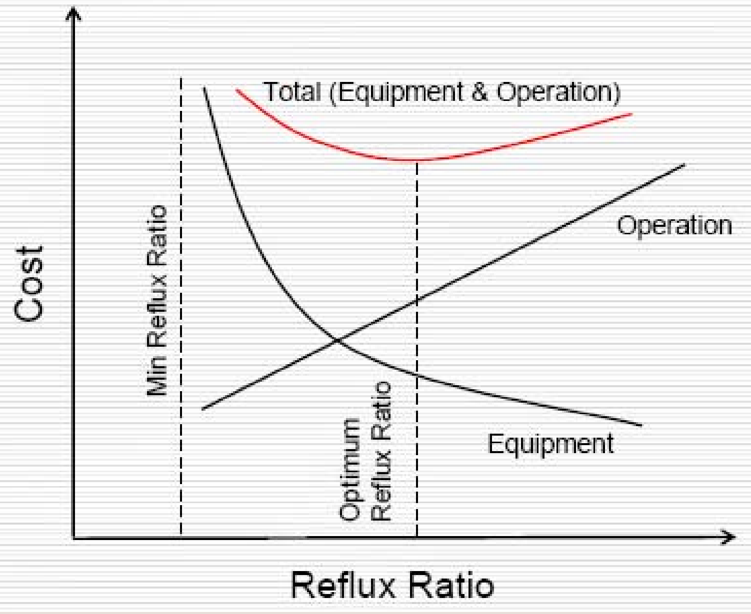
最多理論板數 vs. 無限大回流比





雖然McCabe法只能用在兩成分系統，但可以了解蒸餾系統有下列原則：

a. 在既定的規格下，蒸餾塔有最低迴流比(RR_{MIN})及最少理論板數(N_{MIN})
b. 愈容易分離系統，需要理論板數愈少，回流比愈低
c. 理論板數(設備成本)與回流比(操作成本)設計上有trade-off 關係



最少理論板數公式解

Fenske Equation:

$$N_{MIN} = \frac{\log\left[\left(\frac{x_{D,LK}}{x_{D,HK}}\right)\left(\frac{x_{B,HK}}{x_{B,LK}}\right)\right]}{\log(\alpha_{LK/HK})_{avg}}$$

$X_{D,LK}$: 塔頂出料 Light key 濃度
$X_{D,HK}$: 塔頂出料 Heavy key 濃度
$\alpha_{LK/HK}$: Light key / Heavy key 相對揮發度

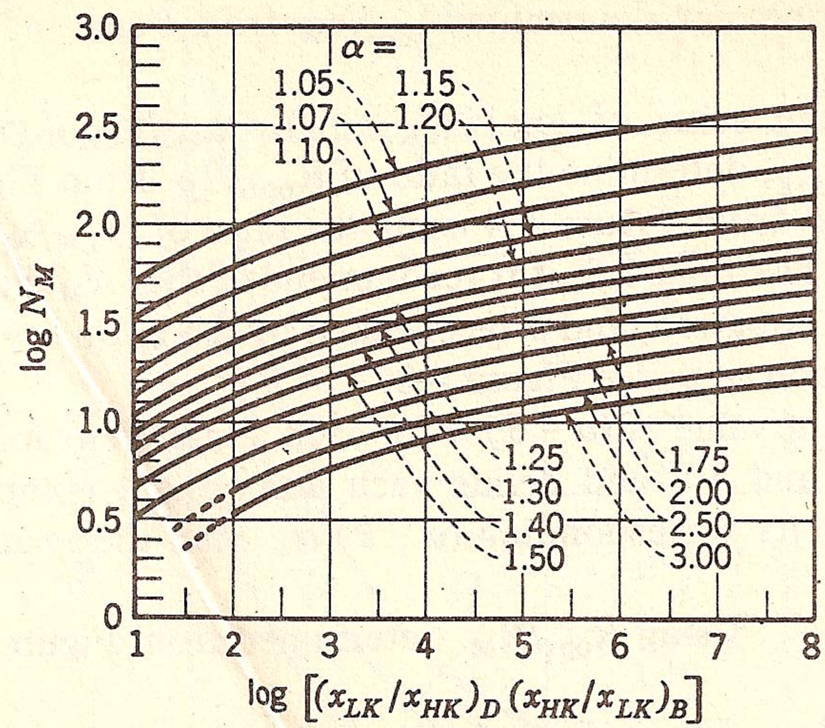
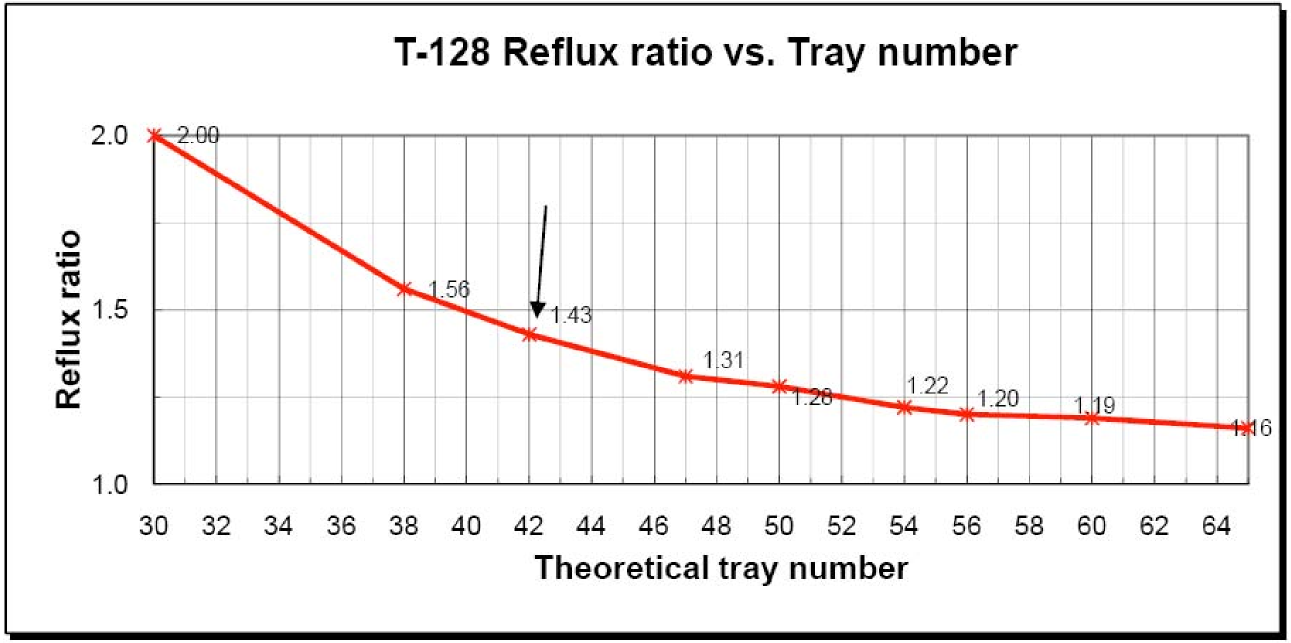


Figure 4.1.(1) Fenske equation for minimum plates expressed in graph form.

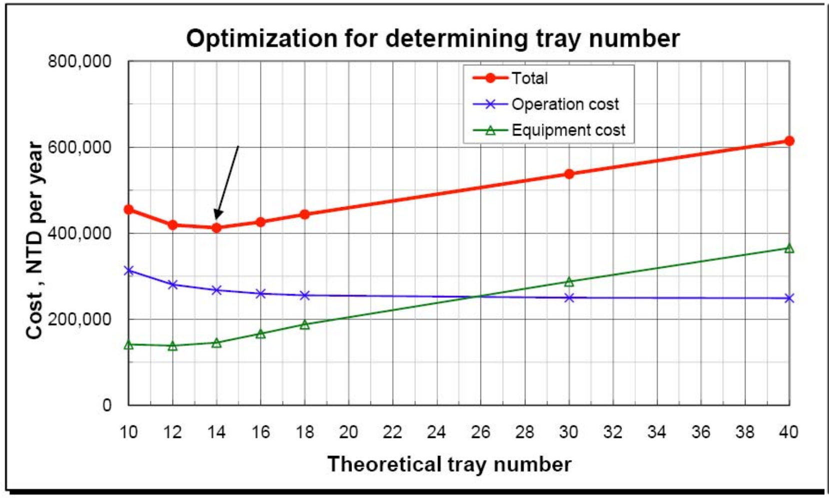
蒸餾塔經驗法設計

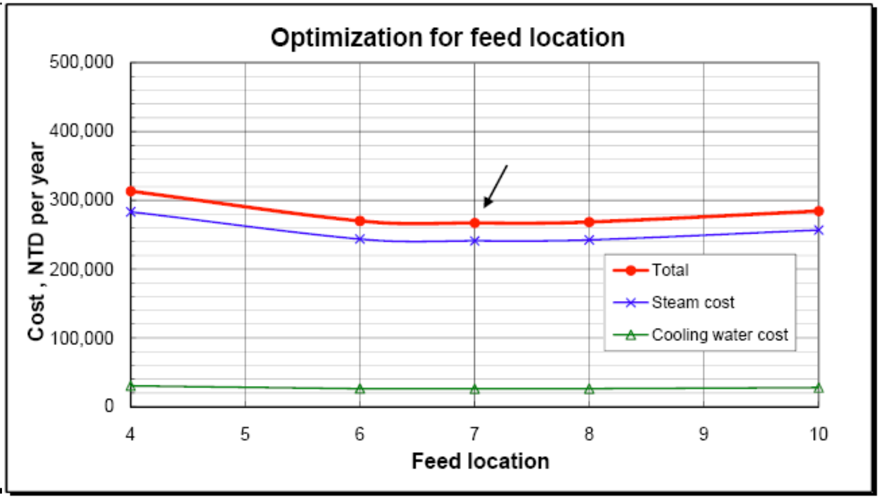
一般經驗法則，回流比設計在R_{MIN}的1.1 倍或理論板數設計在N_{MIN}的2倍



蒸餾塔最佳化設計

1. 設定塔頂與塔底規格進行模擬
2. 改變不同理論板、進料板位置
3. 畫出設備成本(<u>塔體及塔內件成本</u>)與操作成本(<u>蒸氣與冷卻水費用</u>)關係圖





蒸餾塔模擬結果可提供資訊

a. 理論板數 (利用板效率或HETP數據可算出實際板數或填料高度)
b. 進出口流量與組成規格
c. 再沸器與冷凝器能耗
d. 塔溫與塔壓力變化 (須搭配實際板數或填料高度與塔徑結果)
e. 塔內水力數據
- 氣液相流量
- 塔內組成變化
- 塔內氣液相密度、黏度、分子量、液體表面張力

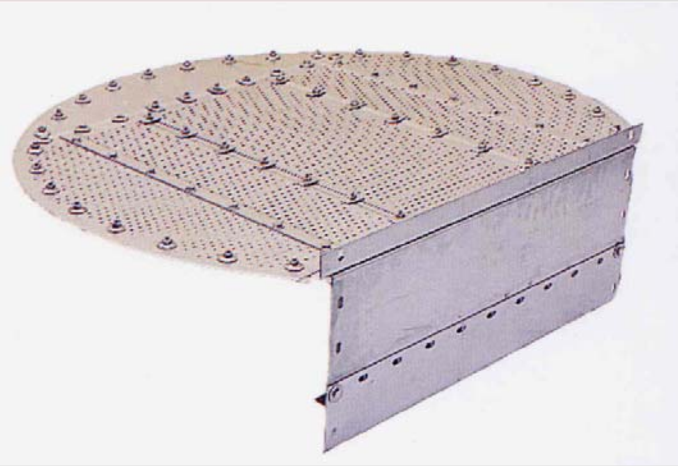
蒸餾塔塔內水力數據 – Vapor phase

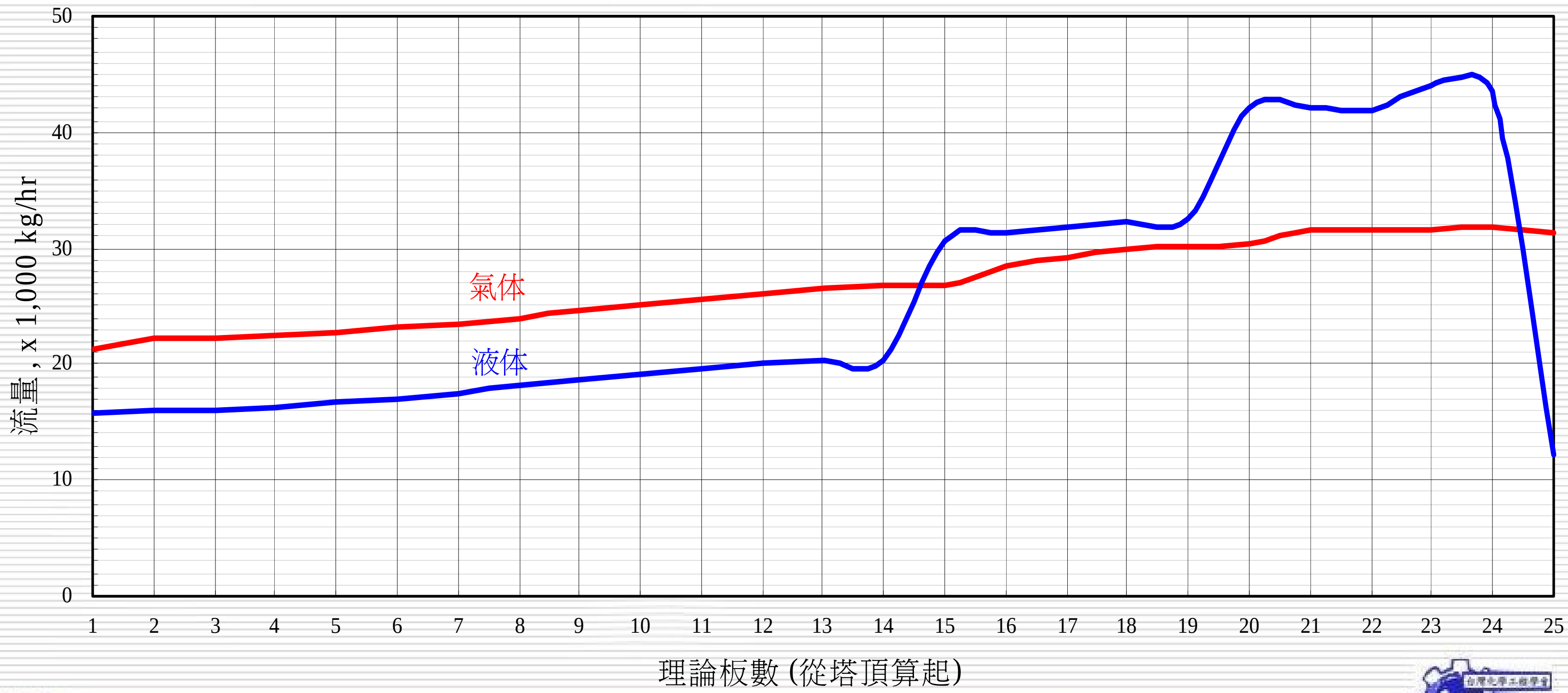
Tray	Temp deg C	Pressure KG/CM2	Molec Wt	Rate K*KG/HR	Density at Cond KG/M3	Viscosity at Cond CP
	------------ Vapor to Tray ------------					
1	100.06	0.2240000	117.5539	1.9544204	0.841737	0.0076859
2	109.88	0.2393846	126.8317	2.1392858	0.946963	0.0075239
3	113.49	0.2547692	129.3747	2.2049863	1.019348	0.0074984
4	115.51	0.2701538	129.8736	2.2191758	1.080197	0.0075186
5	117.15	0.2855385	129.9423	2.2190922	1.138246	0.0075476
6	118.69	0.3009231	129.9258	2.2157421	1.195497	0.0075774
7	120.17	0.3163077	129.8914	2.2111984	1.252336	0.0076069
8	121.64	0.3316923	129.8509	2.2054794	1.308782	0.0076360
9	123.12	0.3470769	129.8045	2.1979808	1.364728	0.0076652
10	124.65	0.3624615	129.7491	2.1876947	1.419980	0.0076951
11	126.29	0.3778462	129.6799	2.1732510	1.474245	0.0077268
12	134.63	0.3932308	138.8012	2.8083736	1.611713	0.0075326
13	139.12	0.4086154	141.4593	2.8876626	1.690032	0.0075037
14	143.54	0.4240000	142.0766	2.8257270	1.743998	0.0075358
15	Reboiler					

蒸餾塔塔內水力數據 – Liquid phase

Tray	Temp deg C	Molec Wt	Rate K*KG/HR	Density at Cond KG/M3	Viscosity at Cond CP	Surface Tension DYNE/CM
	Liquid from tray					
1	69.54	117.5539	1.9544204	759.66583	0.3826704	19.418010
2	100.06	133.8660	1.2842424	734.97946	0.3110546	17.065127
3	109.88	138.1754	1.3499429	726.82688	0.2926553	16.336972
4	113.49	139.0048	1.3641323	723.65332	0.2850212	16.051952
5	115.51	139.1334	1.3640488	721.79938	0.2802450	15.883773
6	117.15	139.1269	1.3606987	720.26850	0.2762730	15.745207
7	118.69	139.0956	1.3561550	718.85208	0.2726947	15.618423
8	120.17	139.0615	1.3504360	717.50719	0.2694421	15.500293
9	121.64	139.0294	1.3429374	716.22549	0.2665335	15.391096
10	123.12	139.0012	1.3326513	715.00777	0.2640311	15.292449
11	124.65	138.9788	1.3182076	713.85889	0.2620383	15.206988
12	126.29	138.9646	2.9533310	712.78555	0.2606925	15.138042
13	134.63	141.4950	3.0326200	706.46778	0.2526228	14.651624
14	139.12	142.0831	2.9706843	704.08244	0.2524216	14.530793
15	143.54	142.2098	0.1449574	702.76288	0.2560455	14.537027

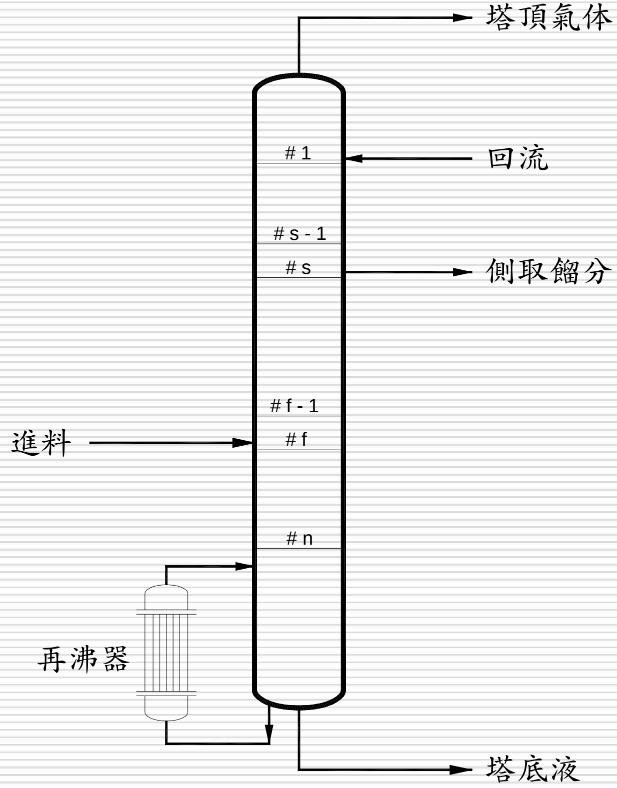
利用模擬軟體產生塔內各板氣体、液体流量作為塔徑計算之用



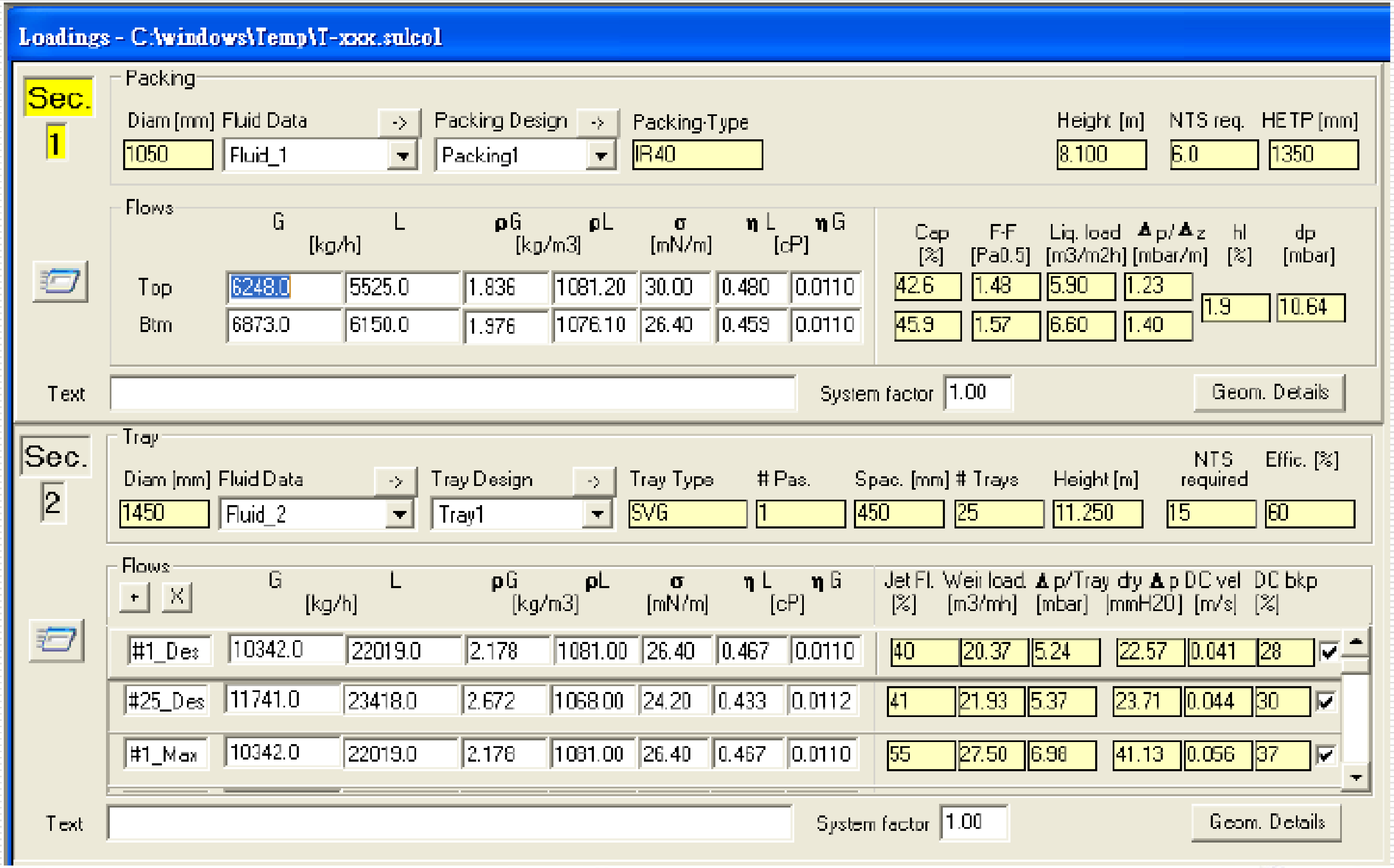


蒸餾塔塔徑與塔內件計算

- 將模擬得到的塔內部水力數據輸入塔內件廠商設計軟體，進行塔徑計算與塔內件形式的選擇。
- 下列位置作分段計算
  a. 塔進料位置
  b. 塔頂塔底出料
  c. 使用不同形式內件Packing或Tray分界
  d. 塔側邊氣相或液相取出位置
  e. Recycle 或 Pumping around 回塔位置
  f. 氣液量負荷最大位置



蒸餾塔徑與塔內件型式計算 (by SULCOL software)



	G [kg/h]	L	ρG [kg/m3]	ρL	σ [mN/m]	η L [cP]	η G	Cap [%]	F-F [Pa0.5]	Liq. load [m3/m2h]	Δp/Δz [mbar/m]	hl [%]	dp [mbar]
Top	6248.0	5525.0	1.836	1081.20	30.00	0.480	0.0110	42.6	1.48	5.90	1.23	1.9	10.64
Btm	6873.0	6150.0	1.976	1076.10	26.40	0.459	0.0110	45.9	1.57	6.60	1.40		

	G [kg/h]	L	ρG [kg/m3]	ρL	σ [mN/m]	η L [cP]	η G	Jet Fl. [%]	Weir load [m3/mh]	Δp/Tray [mbar]	dry Δp [mmH2O]	DC vel [m/s]	DC bkp [%]
#1_Des	10342.0	22019.0	2.178	1081.00	26.40	0.467	0.0110	40	20.37	5.24	22.57	0.041	28
#25_Des	11741.0	23418.0	2.672	1068.00	24.20	0.433	0.0112	41	21.93	5.37	23.71	0.044	30
#1_Max	10342.0	22019.0	2.178	1081.00	26.40	0.467	0.0110	55	27.50	6.98	41.13	0.056	37

蒸餾塔規格書

CCP 長春石油化學 / 長春人造樹脂 / DCC 大連化學工業	Specification Sheet Tower Sheet No : 1 of 2	☑ 長春石油化學 ☐ 長春人造樹脂 ☐ 大連化工公司 ☐ 長連化工公司	Project : xxx
		☐ 新竹廠 ☑ 苗栗廠 ☐ 麥寮廠 ☐ 高雄廠 ☐ 大發廠 ☐ 盤錦廠	

1	Equipment No. :	T-xxx	Equipment Name :	Light Ends Column					
2	No. Required :	1	Type :	☑ Plate	☐ Structured Packing		☑ Random Packing		
3	External Streams		Feed	Reflux	Overhead	Distillate	Distillate	Bottom	
4					Vapor	Oil Phase	Water Phase	Liquid	

OPERATING CONDITIONS

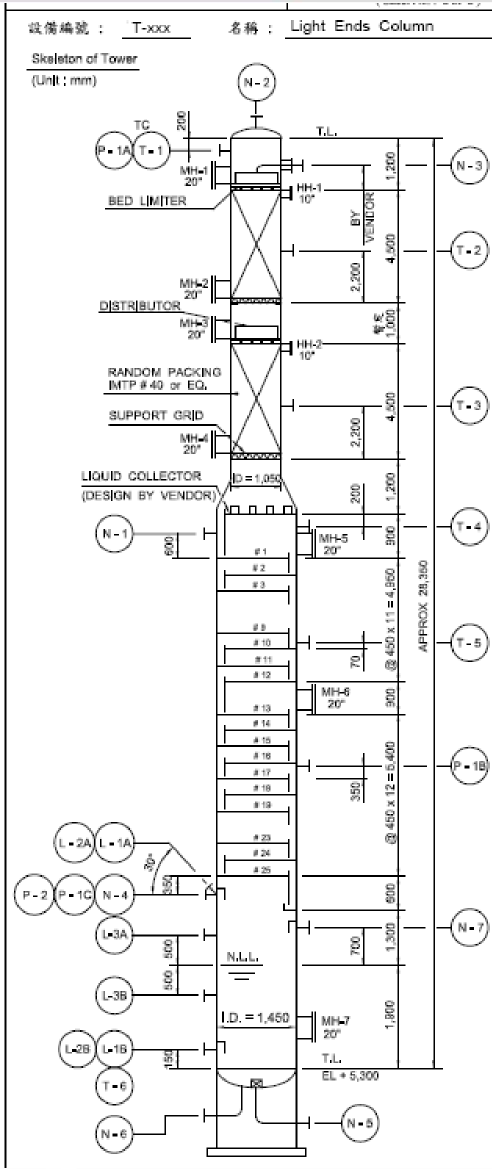
			Feed	Reflux	Overhead Vapor	Distillate Oil Phase	Distillate Water Phase	Bottom Liquid
5	Position		Tray #1	Top	Top	Top	Top	Bottom
6	Phase		Liquid	Liquid	Vapor	Liquid	Liquid	Liquid
7	Flow Rate	kg/h	12,400	4,007	4,731	534	175	11,677
8	Pressure	torr	563	550	550	550	550	668
9	Temperature	°C	45	40	92	40	40	115.6
10	Density	kg/m^3	1156.8	1092.4	1.550	1092.4	992.4	1071.5
11	Viscosity	cp	0.870	0.720	0.011	0.720	0.670	0.438

	Internal Condition (for tower sizing)		Packing Section top Liquid	Packing Section top Vapor	Packing Section bottom Liquid	Packing Section bottom Vapor	Tray Section top Liquid	Tray Section top Vapor	Tray Section bottom Liquid	Tray Section bottom Vapor
15	Flow Rate	kg/h	5,525	6,248	6,150	6,873	22,019	10,342	23,418	11,741
16	Pressure	torr	550		560		563		668	
17	Temperature	°C	100.1		103.7		107.6		115.6	
18	Density	kg/m^3	1,081.2	1.836	1,076.1	1.976	1,081.0	2.178	1,068.0	2.672
19	Viscosity	cp	0.480	0.011	0.459	0.011	0.467	0.011	0.433	0.011
20	Surface Tension	dyne/cm	30.0		26.4		26.4		24.2	
21	Solid content	kg/h								

蒸餾塔規格書(續)

22	DESIGN	Code / Law		ASME SEC.VIII Div.1	Code Stamp		☐ Yes ☑ No
23		Design Press.	kg/cm^2G	1.9; Full Vacuum	Radiograph		Spot
24		Design Temp.	°C	150	Stress Relief		
25		Corrosion Allowance	mm	0	Test Pressure (Hydr. / Pneum.)	kg/cm^2G	
26							
27	CONSTRUCTION	Shell ID	mm	$1,050^{Upper}$ / $1,450^{Lower}$	Packing Type & Size		IMTP #40 or EQ.
28		Shell Height (T.L. to T.L.)	mm	Approx. 28,350	Packing Height	mm	4,500 x 2
29		Shell Thickness	mm		Distributor Type		by Vendor
30		Skirt Height	mm	5,000	Redistributor Type		by Vendor
31		Head Type		1:2 Ellip	Demister Type		
32		Tray Type		Fixed valve	Insulation	mm	150
33		Tray Number		25	Fire Proof	mm	Yes
34		Tray Spacing	mm	450	Shell Empty Weight	kg	
35		Tray Pass No.		1	Internal Weight	kg	
36	MATERIAL	Shell		SS316L	Bolts & Nuts (**Internal**)		SS316L
37		Head		SS316L	Tray		SS316L
38		Skirt		CS	Packing		SS316L
39		Nozzle		SS316L	Packing Support		SS316L
40		Flange		SS316L	Distributor		SS316L
41		Gasket		PTFE Based	Demister		
42		Bolts & Nuts (**External**)			Other Internals		SS316L
43	REMARKS	Notes:					
44		1. Allowable pressure drop for the whole column should be within 120 mmHg @100% operating load.					
45		2. Due to the upper column is prone to fouling, the distributors should possess anti-fouling feature.					
46							
47							
48							

蒸餾塔規格書(續)



MARK	SIZE	RATING	個數	用途
MH-1~7	20"	ANSI CLASS 150	7	MAN HOLE (W/ DAVIT)
HH-1~2	10"	ANSI CLASS 150	2	HAND HOLE (W/ BAFFLE)
T-1~6	1½"	ANSI CLASS 150	6	T I , TC
P-2	1"	ANSI CLASS 150	1	PG
P-1A~C	2"	ANSI CLASS 150	3	Pd
L-3 A/B	1"	ANSI CLASS 150	2	LG
L-2 A/B	3"	ANSI CLASS 150	2	L I (W/ BAFFLE)
L-1 A/B	3"	ANSI CLASS 150	2	LC (W/ BAFFLE)
N-7	20"	ANSI CLASS 150	1	FROM REBOILER RETURN (W/ BAFFLE)
N-6	14"	ANSI CLASS 150	1	TO REBOILER
N-5	4"	ANSI CLASS 150	1	BOTTOM LIQUID OUTLET (W/ VORTEX BREAKER)
N-4	1½"	ANSI CLASS 150	1	PUMP RD RETURN
N-3	4"/1½"	ANSI CLASS 150	1	REFLUX INLET
N-2	14"	ANSI CLASS 150	1	TOP VAPOR OUTLET
N-1	2½"	ANSI CLASS 150	1	LIQUID FEED

NOZZLE SCHEDULE

- 設備最適化 - 換熱器

a. 設計目標
在合理的設計下，提供足夠換熱面積，且壓降及流速均在合理範圍內，避免設備振動或結垢

b. 最適化設計
設備尺寸愈小價格愈低，但需考慮操作成本、安裝與維護等需求

應用模擬軟體設計換熱器

- 製程模擬軟體搭配換熱器設計軟體

  製程模擬軟體

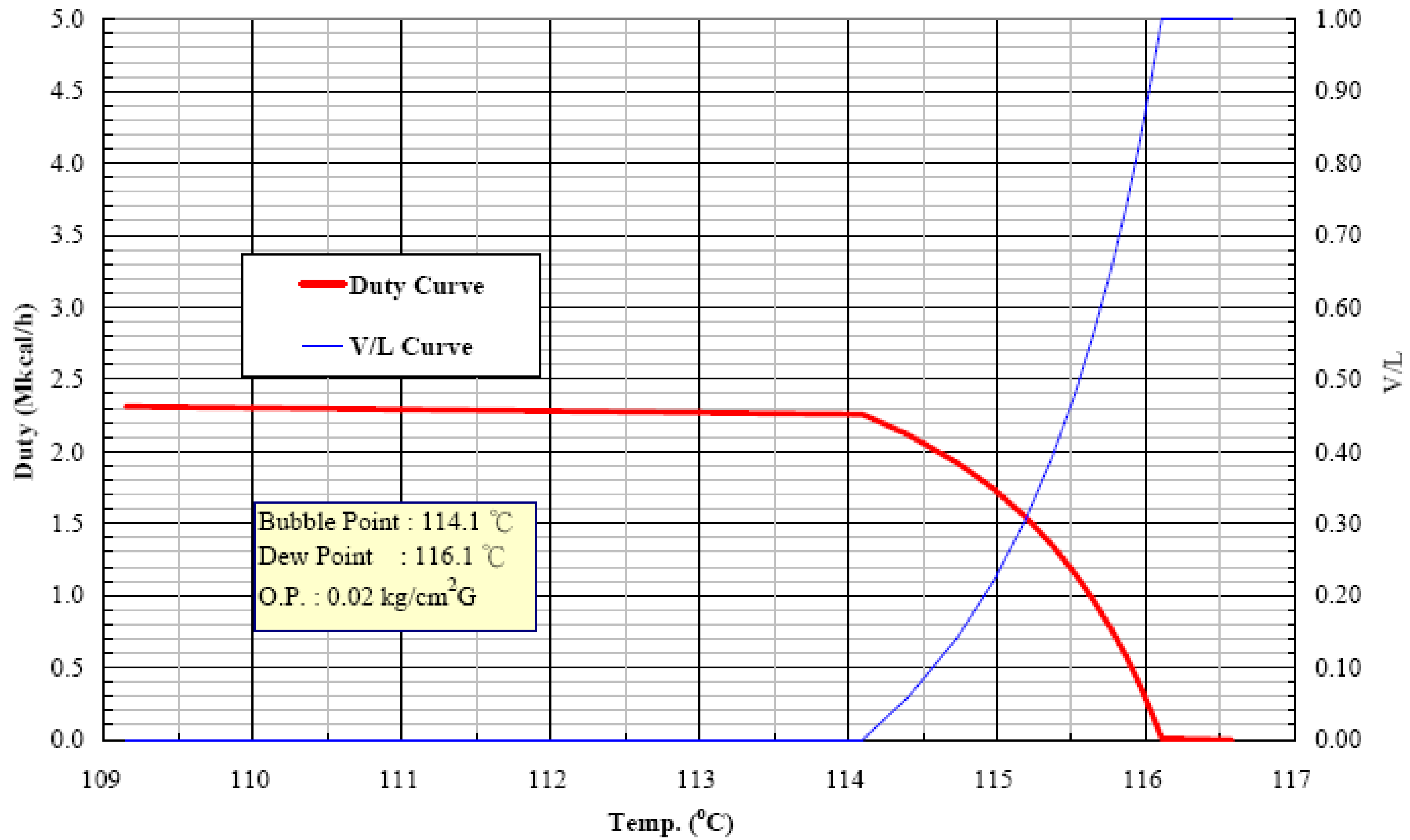
  Aspen Plus or Pro II產生Condensing Curve

  換熱器設計軟體

  HTFS+ (Aspen Tech)、HTRI、

  HEXTRAN (SimSci)

應用模擬軟體設計換熱器 (產生Condensing curve)

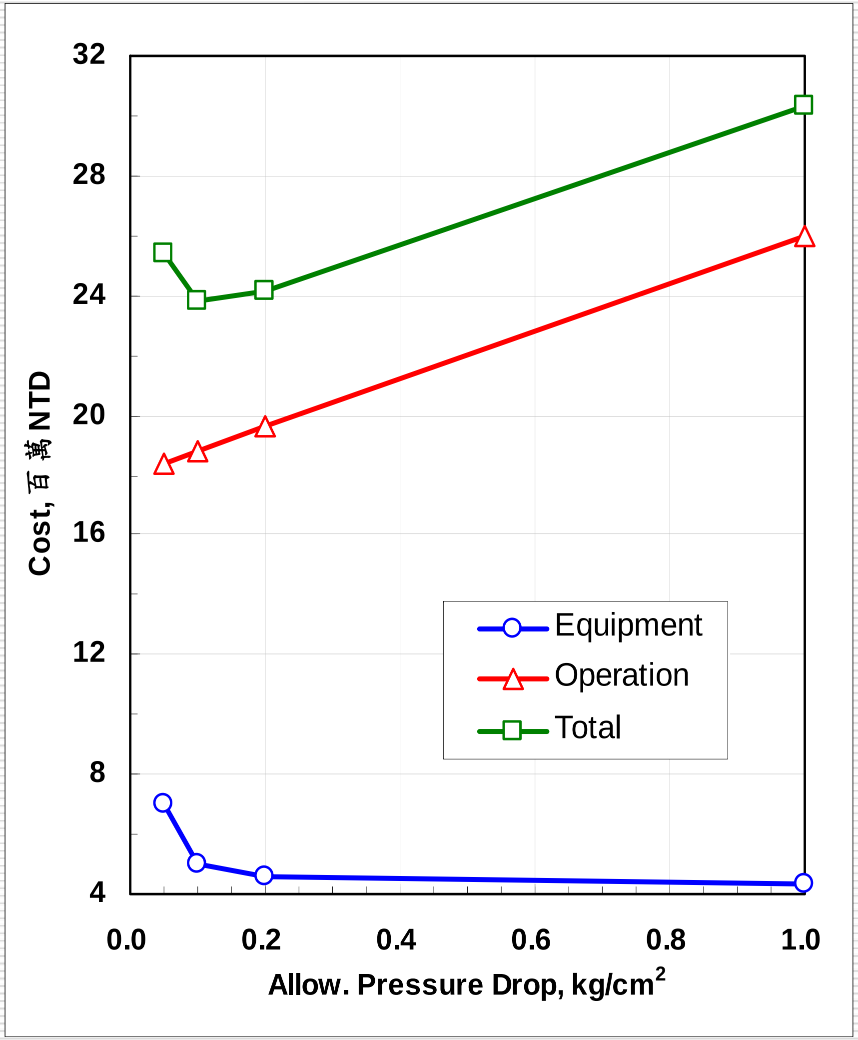


換熱器設計範例

Example: H-123A (T-110 overhead 1st condenser), VAM plant

Allowable Pressure Drop bar	Shell ID mm	Pressure Drop		Baffle		Tube Number (2 pass)	Cost USD
		Shell Side bar	Tube Side bar	Spacing mm	No.		
1	1150	0.615	0.415	248	19	1204	144,880
0.2	1200	0.187	0.344	464	11	1341	152,030
0.1	1250	0.094	0.288	654	7	1488	166,070
0.05	1500	0.048	0.156	692	7	2142	232,710

換熱器設計範例



容許壓降愈大，設備尺寸愈小，設備成本可較低，但壓降大，電力成本變高，操作成本會提高，兩者之間有一最適化之設計。

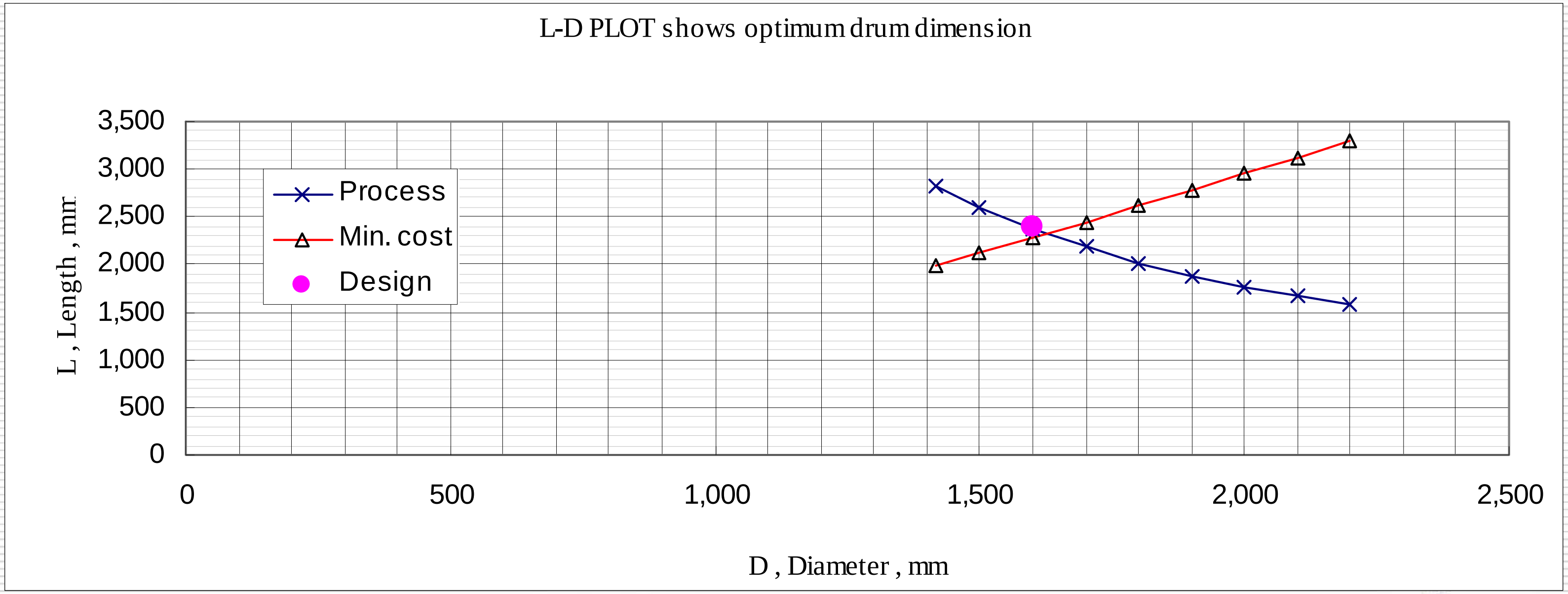
- 設備最適化 - 貯槽

a. 設計目標

設備有足夠的緩衝(或滯留)時間，方便製程操作

b. 最適化設計

設備外徑及高度可調整，但需考慮場地限制及設備成本等因素



五、製程模擬範例:高沸物廢液回收

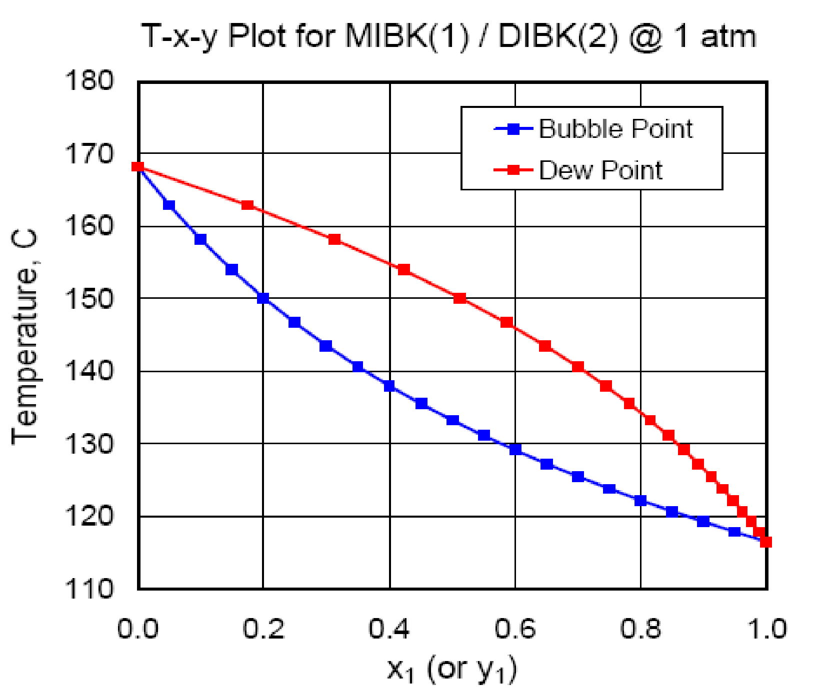
製程高沸物溶液內含MIBK、DIBK及Heavy三種成分，流量為1,000 kg/h，內容物成分如下:

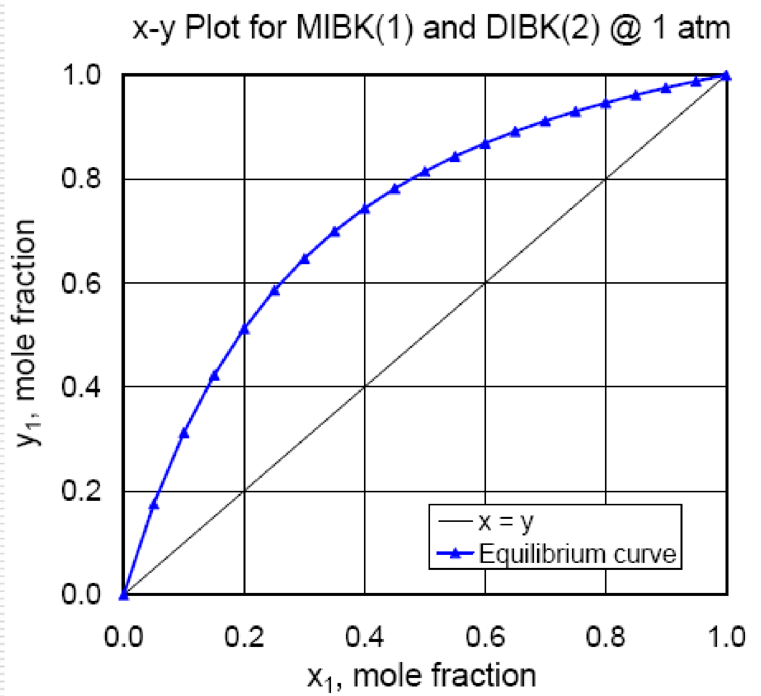
MIBK	52.25 wt%
DIBK	42.75 wt%
Heavy	5.00 wt%

以蒸餾分離方式，分別純化MIBK & DIBK兩項產品，產品純度均須為99 wt%，MIBK & DIBK 的產品回收率則分別為99%及90%以上

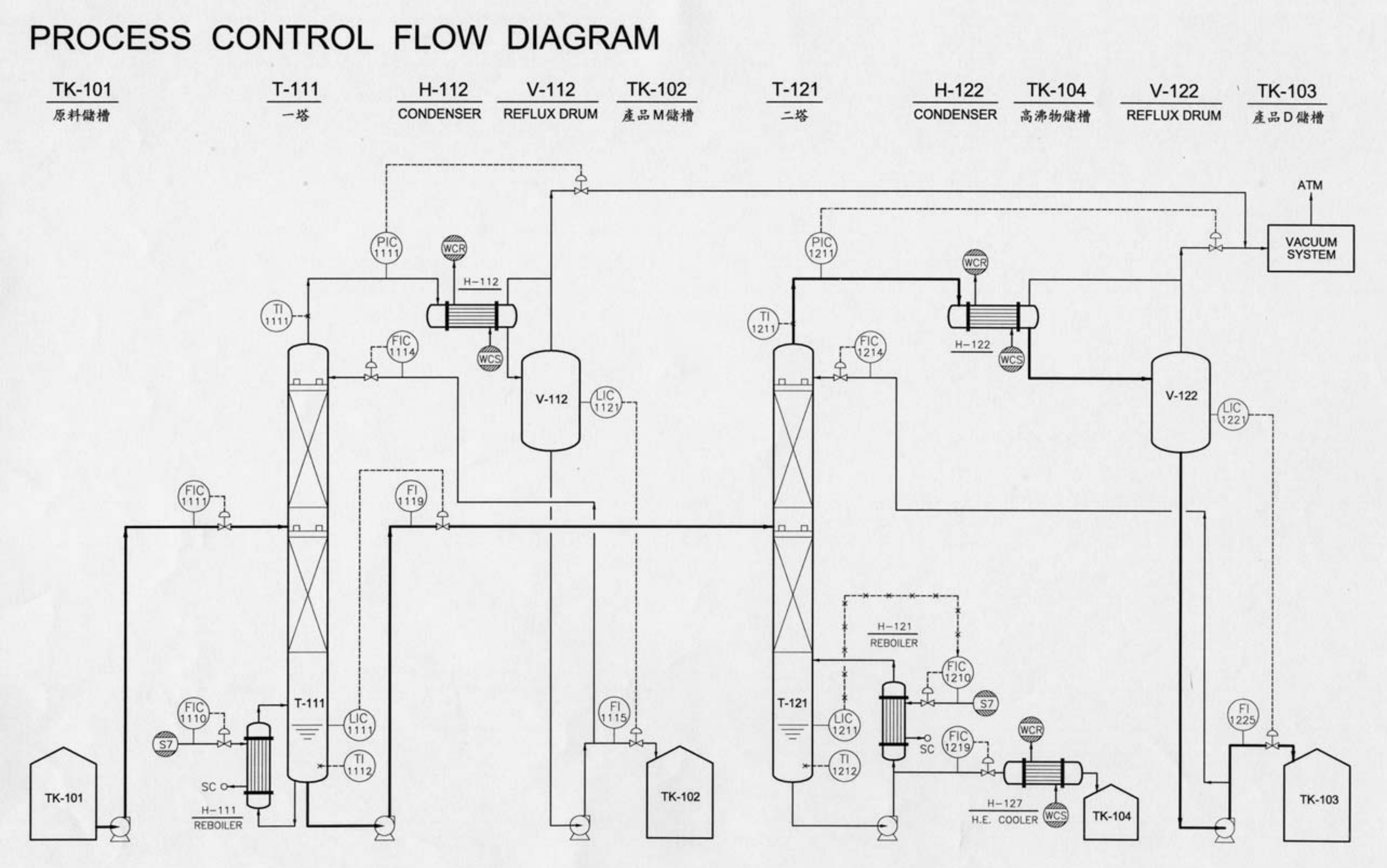
Step 1: 分析 DIBK & MIBK 沸點差異及氣-液相平衡圖

				MW	MP	NBP	Flash point	Auto Ig. Temp.	LEL	UEL	Density @20°C	Azeotrope with Water		Water solubility
					°C	°C	°C	°C	vol%	vol%	kg/m^3	Temp.	mole%	g/100ml
1.	MIBK	**M**ethyl **I**so**b**utyl **K**etone	$C_6H_{12}O$	100.16	-80.0	117.4	14	448	1.2 %	8.0 %	797.8	87.9	90.85%	1.90
2.	DIBK	**D**i**i**so**b**utyl **K**etone	$C_9H_{18}O$	142.24	-46.4	168.0	49	396	0.8 %	6.2 %	847.0			0.05
3.	Heavy	Unknown												





Step 2 : 初步畫出流程圖



Step 3 : 訂定模擬基準

a : 進料條件

MIBK	52.25 wt%
DIBK	42.75 wt%
Heavy	5.00 wt%
流量	1,000 kg/h
溫度	25 ^{o}C

b : 蒸餾塔規格

(i) T1

塔頂	MIBK conc. = 99.0 wt%
塔底	MIBK recovery = 1%

(ii) T2

塔頂	DIBK conc. = 99.0 wt%
塔底	DIBK recovery = 10%

Step 4：選擇適合的模擬軟體，決定操作壓力，並依照Step 2 & 3 資料進行模擬

Step 5：塔板數目及進料位置最適化

Step 6：計算蒸餾塔及相關附屬設備尺寸並開立規格書

Step 7：根據模擬結果擬訂控制策略(含溫控點決定)

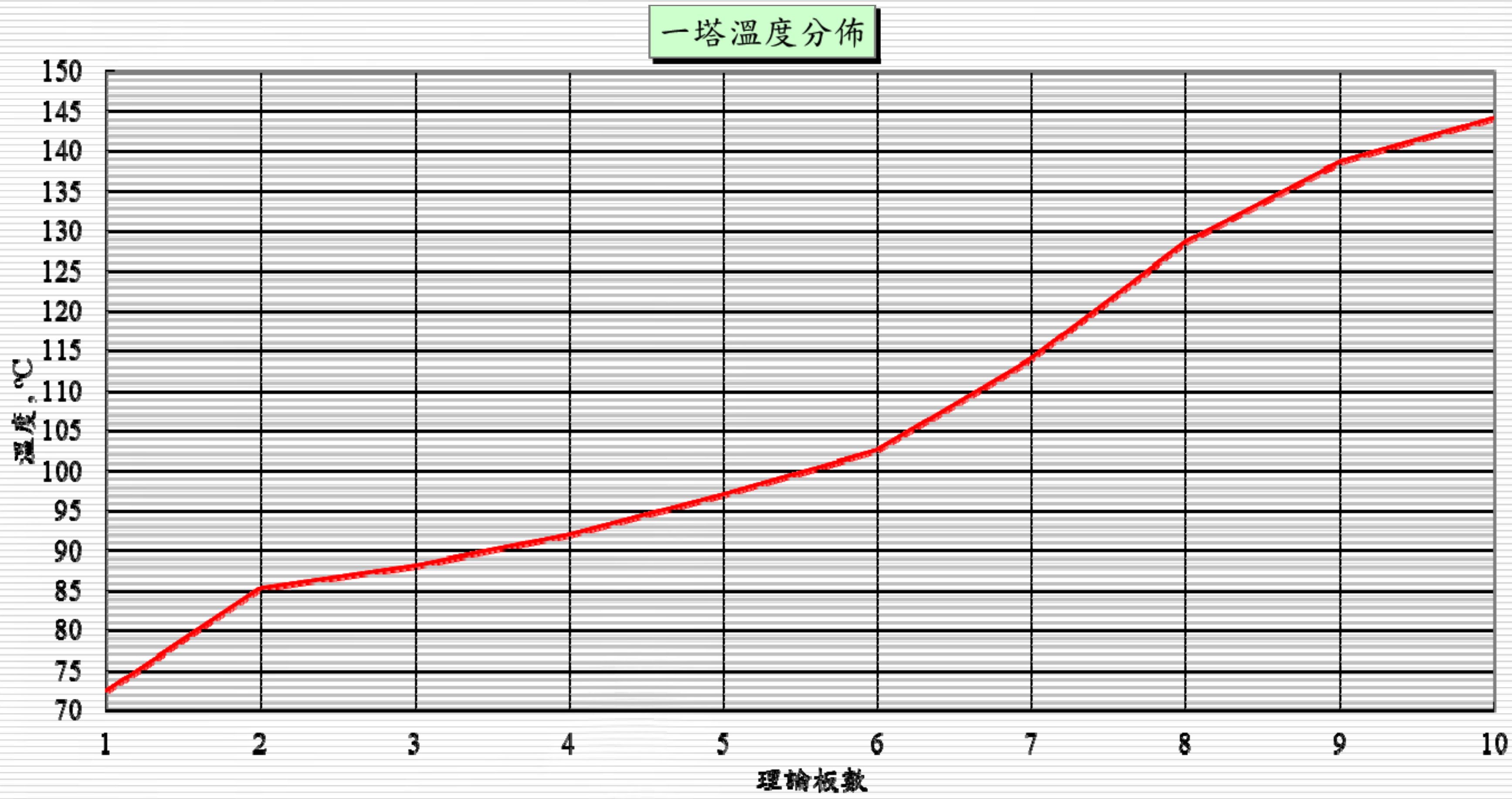
Step 8：繪製 PCF & PID

高沸物廢液回收 - 質量平衡

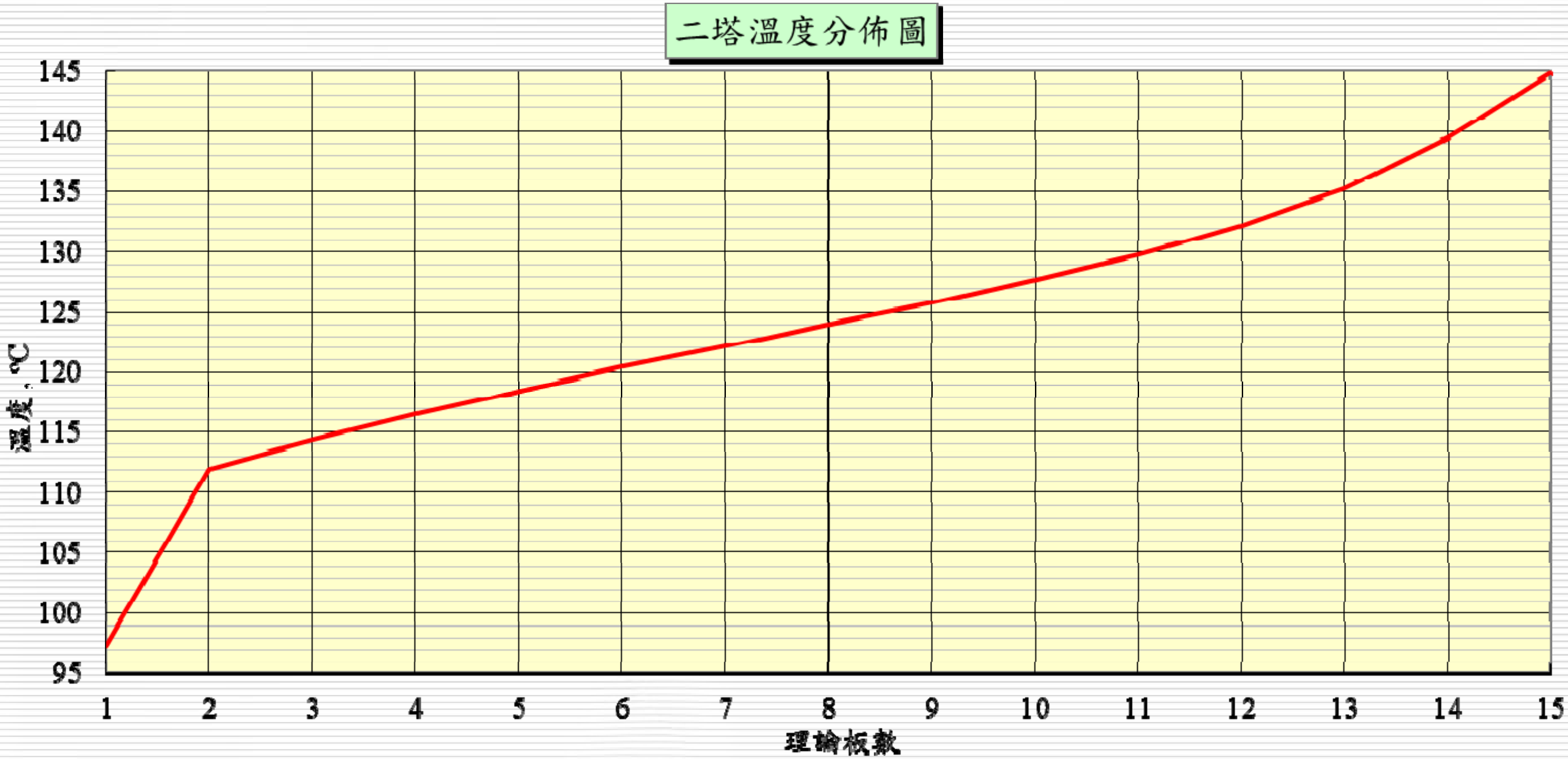
Material Balance

Sream Name		S1	S2	S2r		S3	S4	S4r		S5
Stream Description		一塔 進料	一塔 餾出物	一塔 Reflux	一塔 塔頂氣	一塔 塔底物	二塔 餾出物	二塔 Reflux	二塔 塔頂氣	二塔 塔底物
Phase		液相	液相	液相	氣相	液相	液相	液相	氣相	液相
Total Mass Rate	kg/h	1,000	**524**	321	845	476	**384**	1,707	2,091	**92**
Temerature	^{o}C	25	72.5			144.1	97.4			144.8
Pressure	torr				250				120	
	kg/cm^2A				0.34	0.52			0.163	0.38
Total Weight Comp. Rate	kg/h									
MIBK		522.50	518.66			3.84	3.84			0.00
DIBK		427.50	5.24			422.26	380.03			42.23
Heavy		50.00	0.01			49.99	0.04			49.95
Total Weight Comp. Percents	wt.%									
MIBK		52.25%	99.00%			0.81%	1.00%			0.00%
DIBK		42.75%	1.00%			88.69%	98.99%			45.81%
Heavy		5.00%	0.00%			10.50%	0.01%			54.19%
Density		799.9	752.5		0.845	697.2	739.1		0.805	705.6
Viscosity		0.68	0.33		0.008	0.23	0.34		0.007	0.27
Pipeline Size		1"	3/4"	3/4"	6"	3/4"	3/4"	1"	8"	3/4"

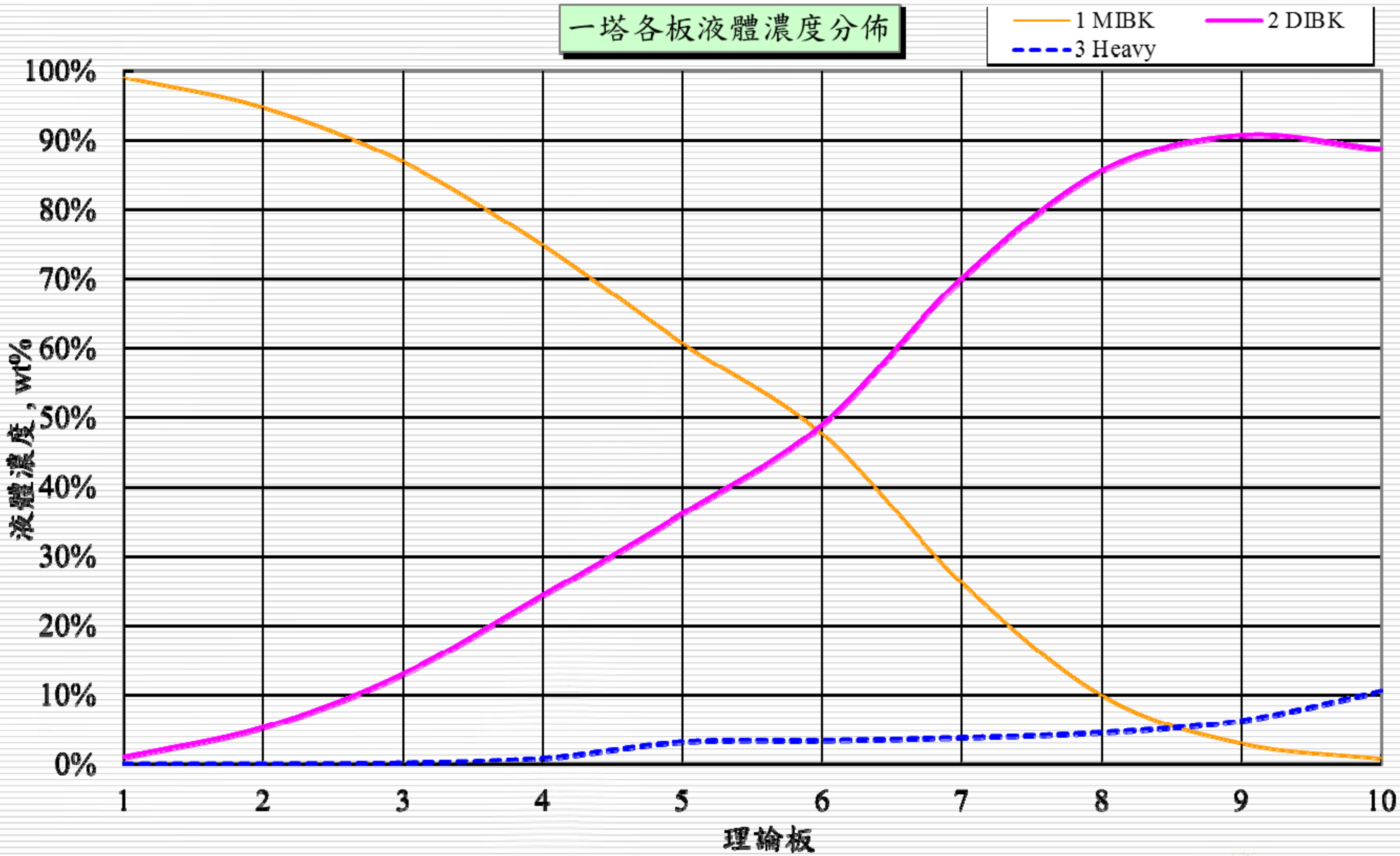
高沸物廢液回收 – T1 塔溫度分佈圖



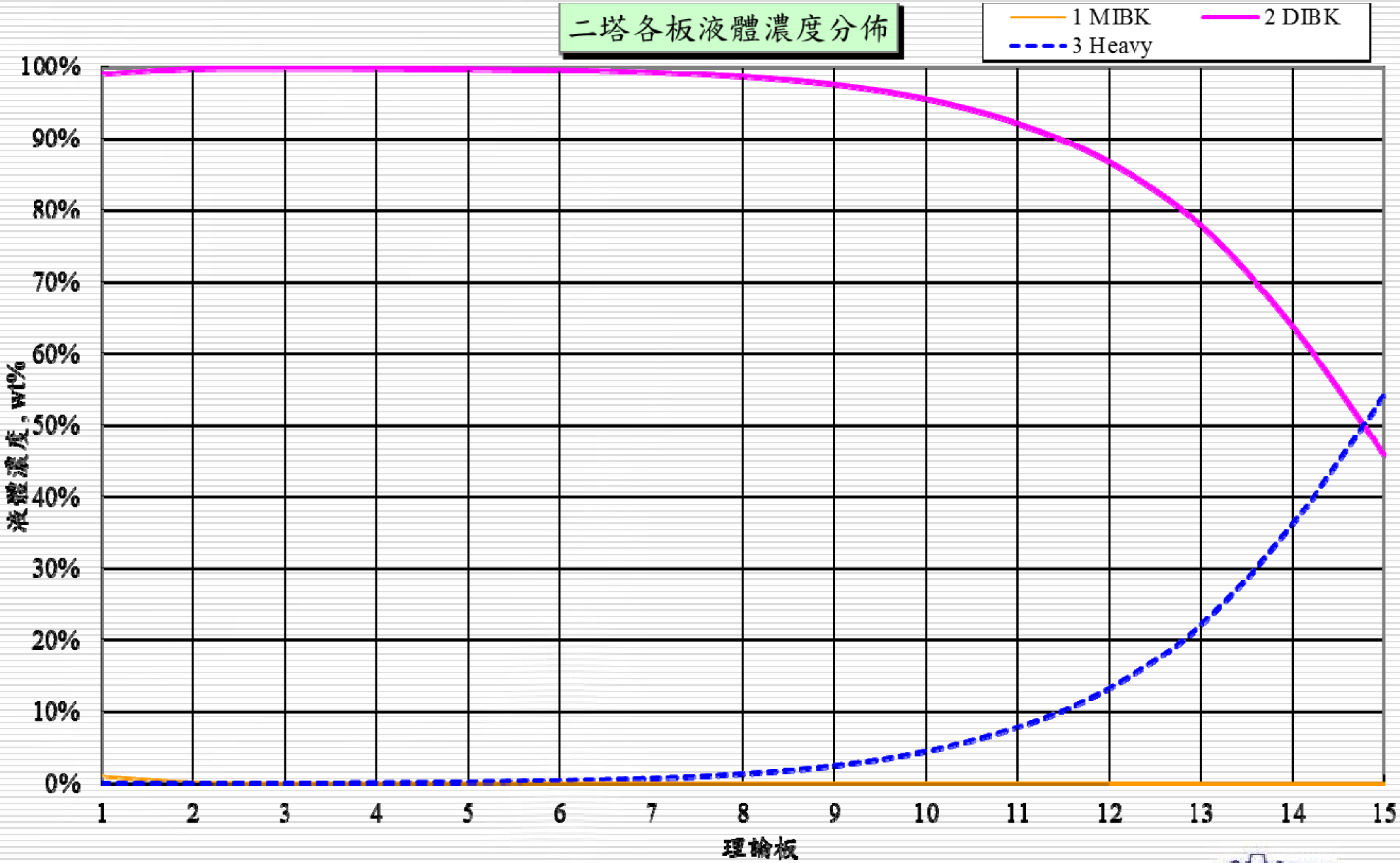
高沸物廢液回收 – T2 塔溫度分佈圖



高沸物廢液回收 – T1 塔液相濃度分佈圖



高沸物廢液回收 – T2 塔液相濃度分佈圖



高沸物廢液回收－模擬結果整理

MIBK	wt%	55 wt%	
DIBK	wt%	45 wt%	
		T-1	**T-2**
Tower Pressure		250 torr	120 torr
Theo. Plates		10	15
Feed Location		5th	12th
Feed Rate	kg/h	**1,000**	476
Overhead		MIBK	DIBK
		99%	99%
	kg/h	**524**	**384**
Bottoms		DIBK(89wt%)	DIBK(46wt%)
		+HE(10.5%)	+HE(54%)
	kg/h	476	**92**
Reflux	kg/h	321	1,707
Reflux Ratio		0.6	4.4
Condenser Duty	kcal/h	83,100	172,900
Cooling Water	m^3/h	8.3	17.3
Reboiler Duty	kcal/h	126,500	162,700
Steam	kg/h	253	325
BTM Vapor	kg/h	1,660	1,947
Vaporization Ratio		3.5	21.2
Temp.--Top	^oC	85	112
--Bottom	^oC	144	145

高沸物廢液回收 – 模擬結果整理(續)

MIBK	wt%	55 wt%	
DIBK	wt%	45 wt%	
Steam cost	NTD/h	126.5	162.5
Cooling Water Cost	NTD/h	5.8	12.1
電力成本	NTD/h		
操作總成本	NTD/h	132.3	174.6
		307	
Tower I.D.	mm	600	800
Packing Height	mm	5,000	10,000
Packing Type		IMTP #25	IMTP #25
Packing Material		SS304	SS304

結論

1. 化工設計通常會有多種組合解，最佳的設計是在合適的操作條件下，設計出操作簡單、價格低廉、操作彈性大的設備，此設計常會因製程差異而有所不同。
2. 藉助於化工設計軟體，可節省計算時間，設計出最適化的規格。

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問題及討論